CSE 4621

Machine Learning

Topic: Practical Advice

Source: DeepLearning.AI

Presented by

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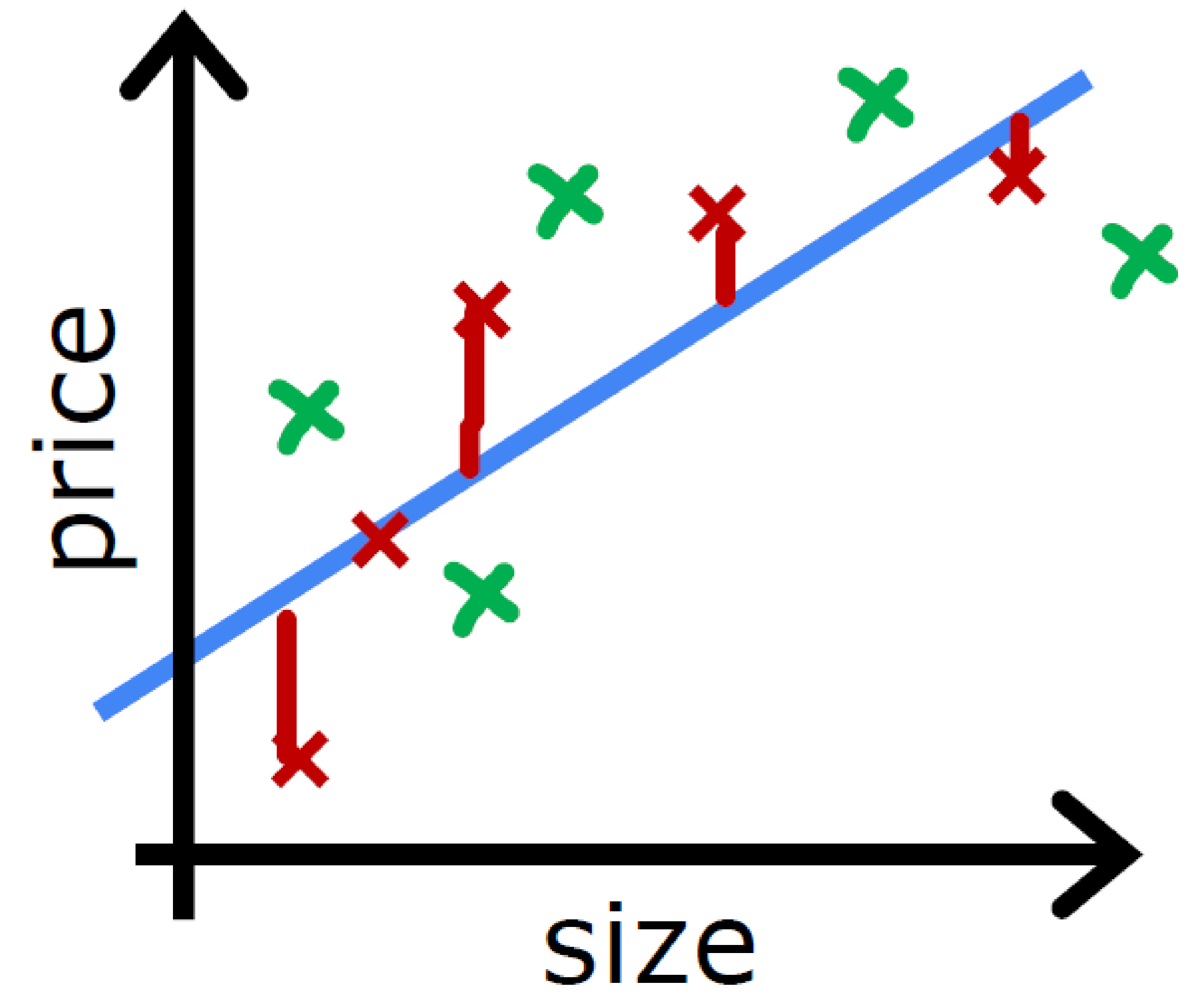
Assistant Professor

Bias/variance

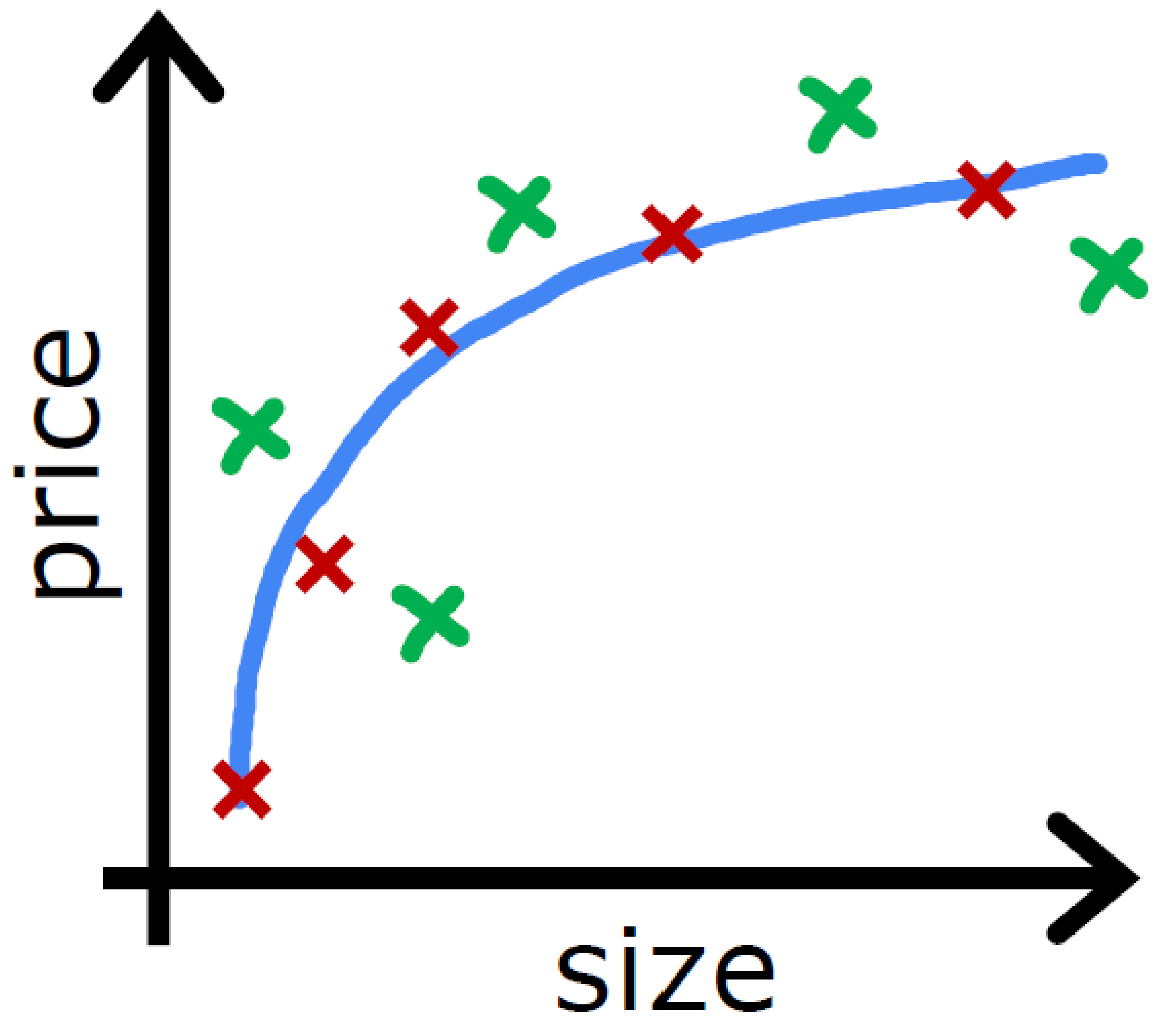


$$f_{\vec{w},b}(x) = w_1 x + b$$

High bias (underfit)

$d = 1$ J_{train} is high

J_{cv} is high

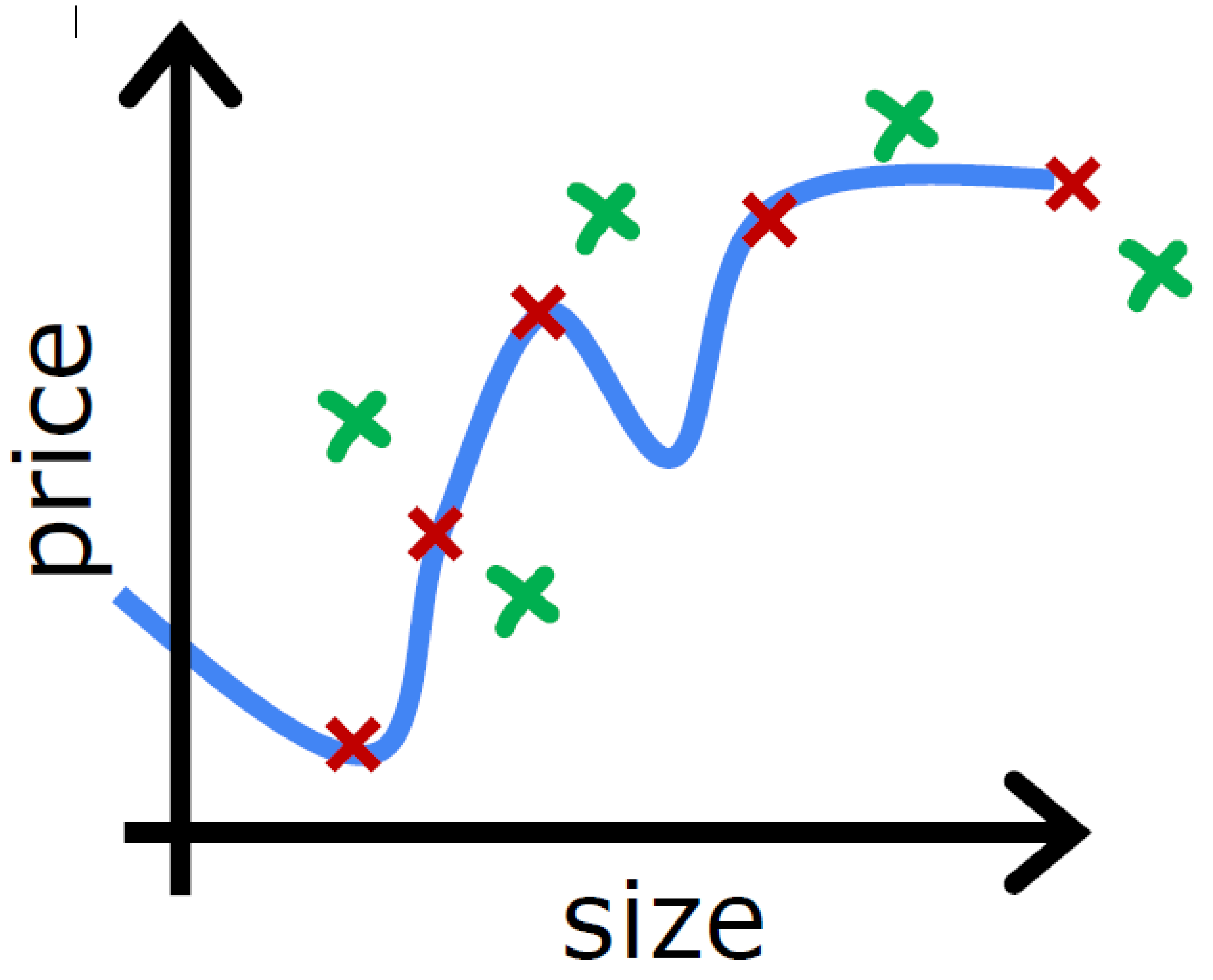


$$f_{\vec{w},b}(x) = w_1 x + w_2 x^2 + b$$

"Just right"

$d = 2$ J_{train} is low

J_{cv} is low



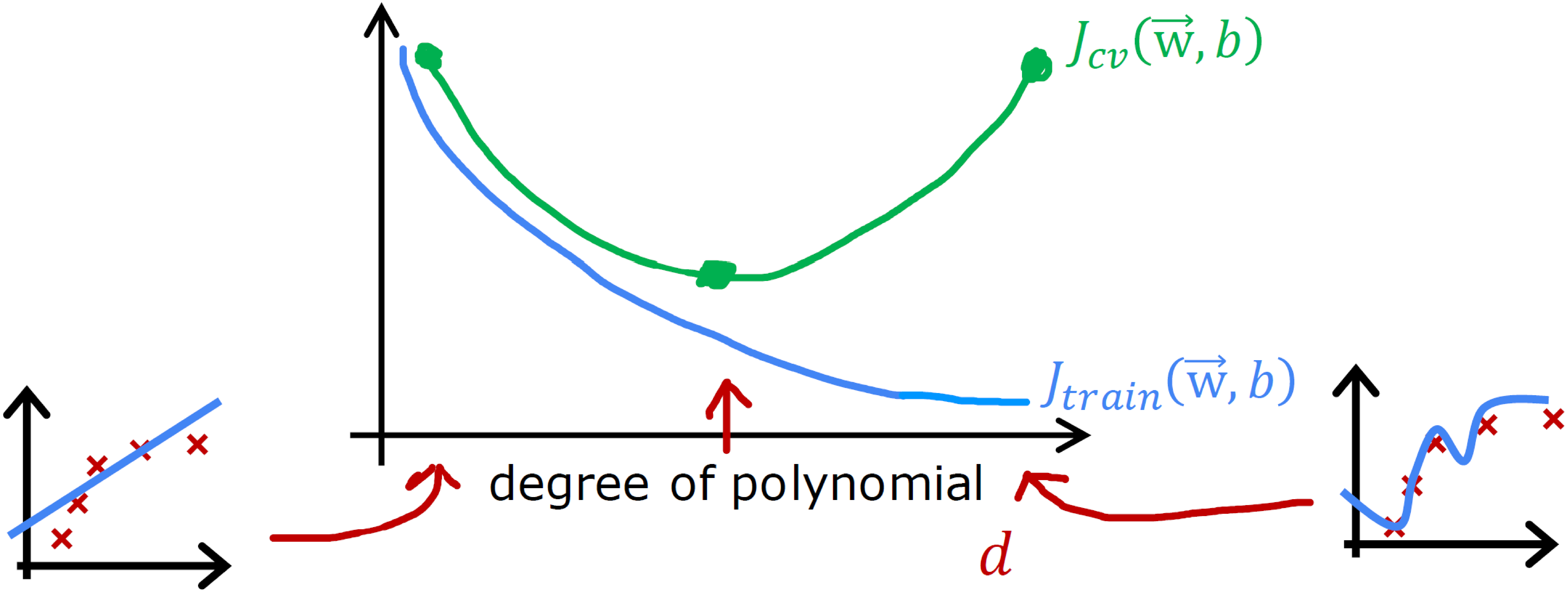
$$f_{\vec{w},b}(x) = w_1 x + w_2 x^2 + w_3 x^3 + w_4 x^4 + b$$

High variance (overfit)

$d = 4$ J_{train} is low

J_{cv} is high

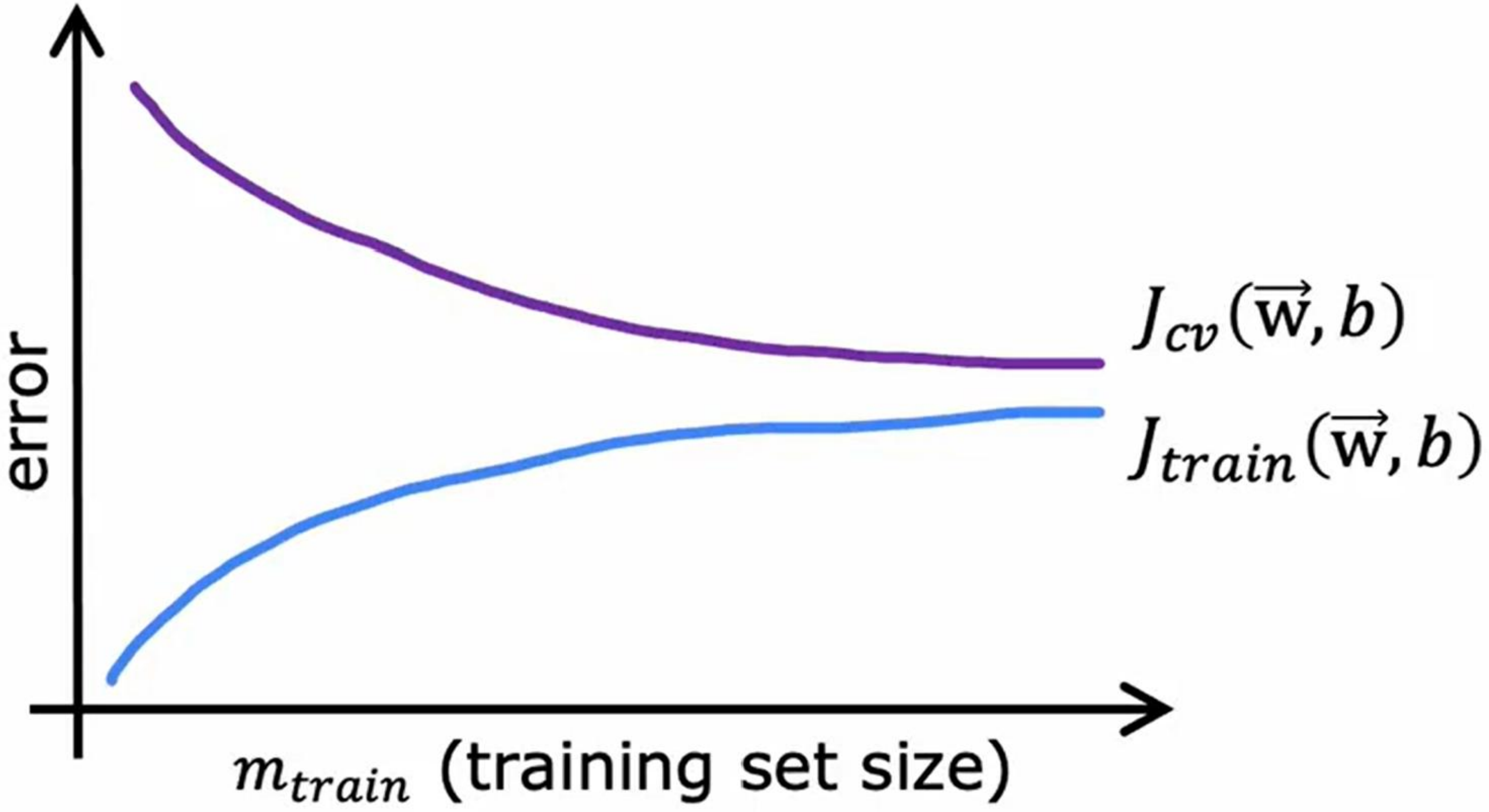
Understanding bias and variance



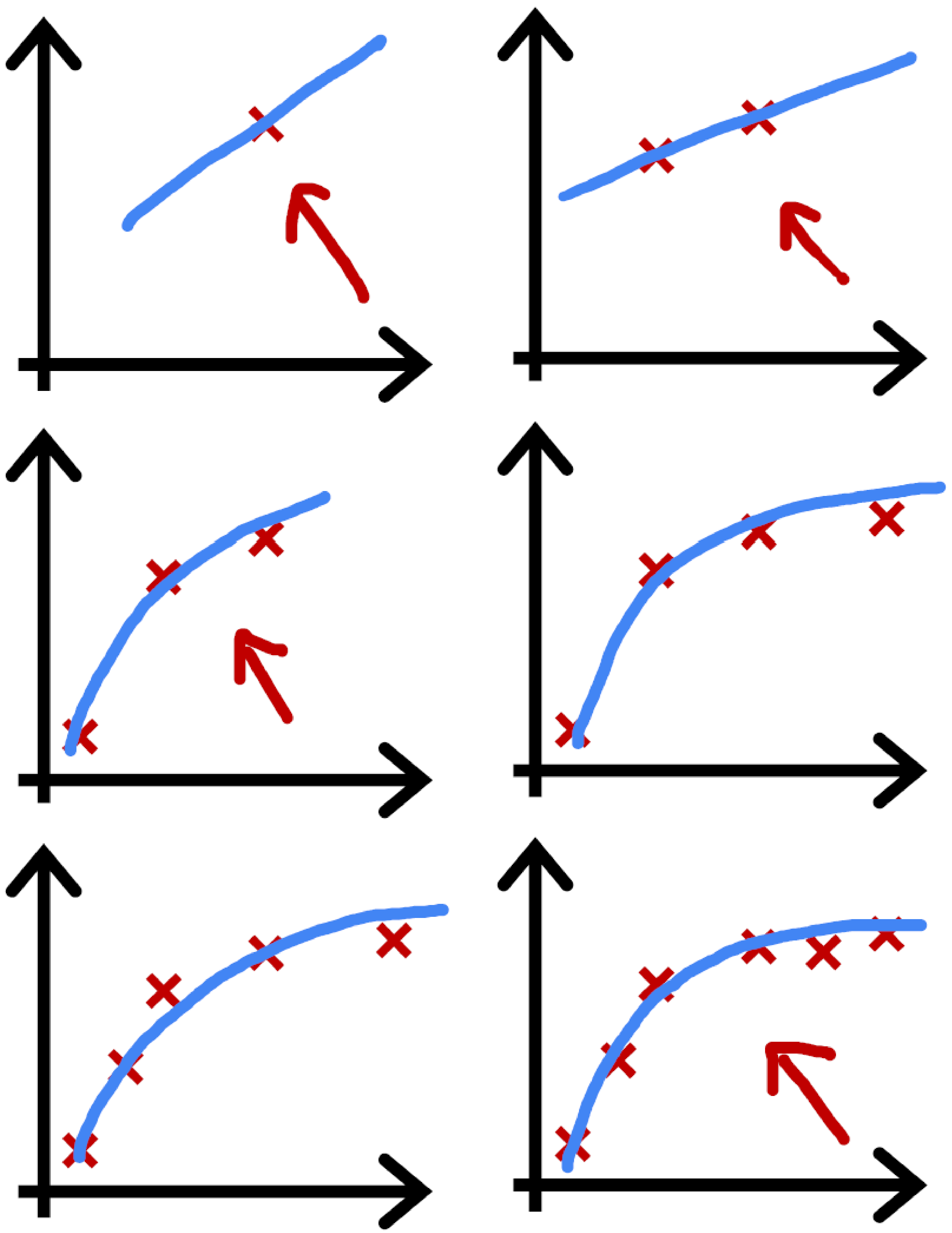
Learning curves

J_{train} = training error

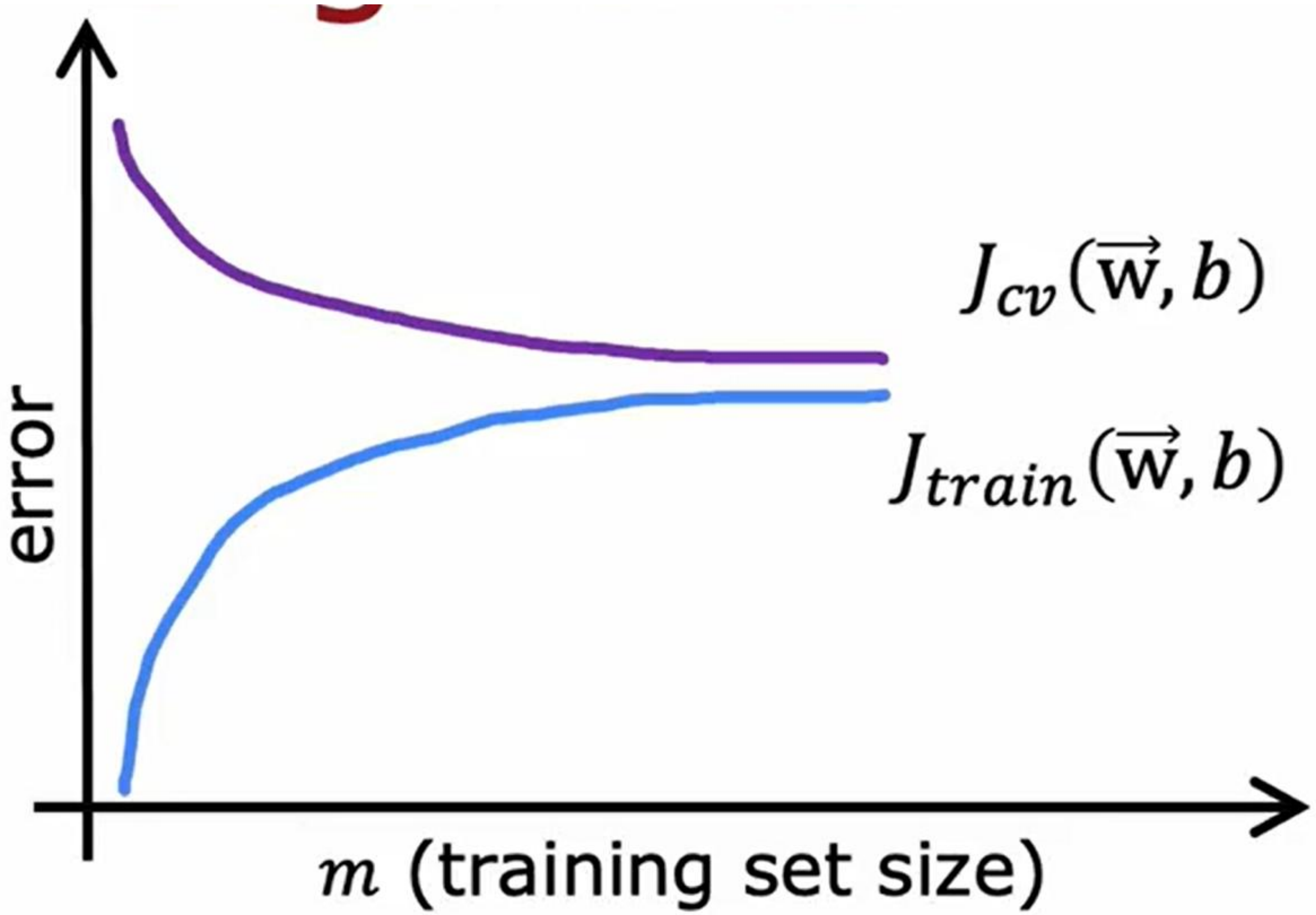
J_{cv} = cross validation error



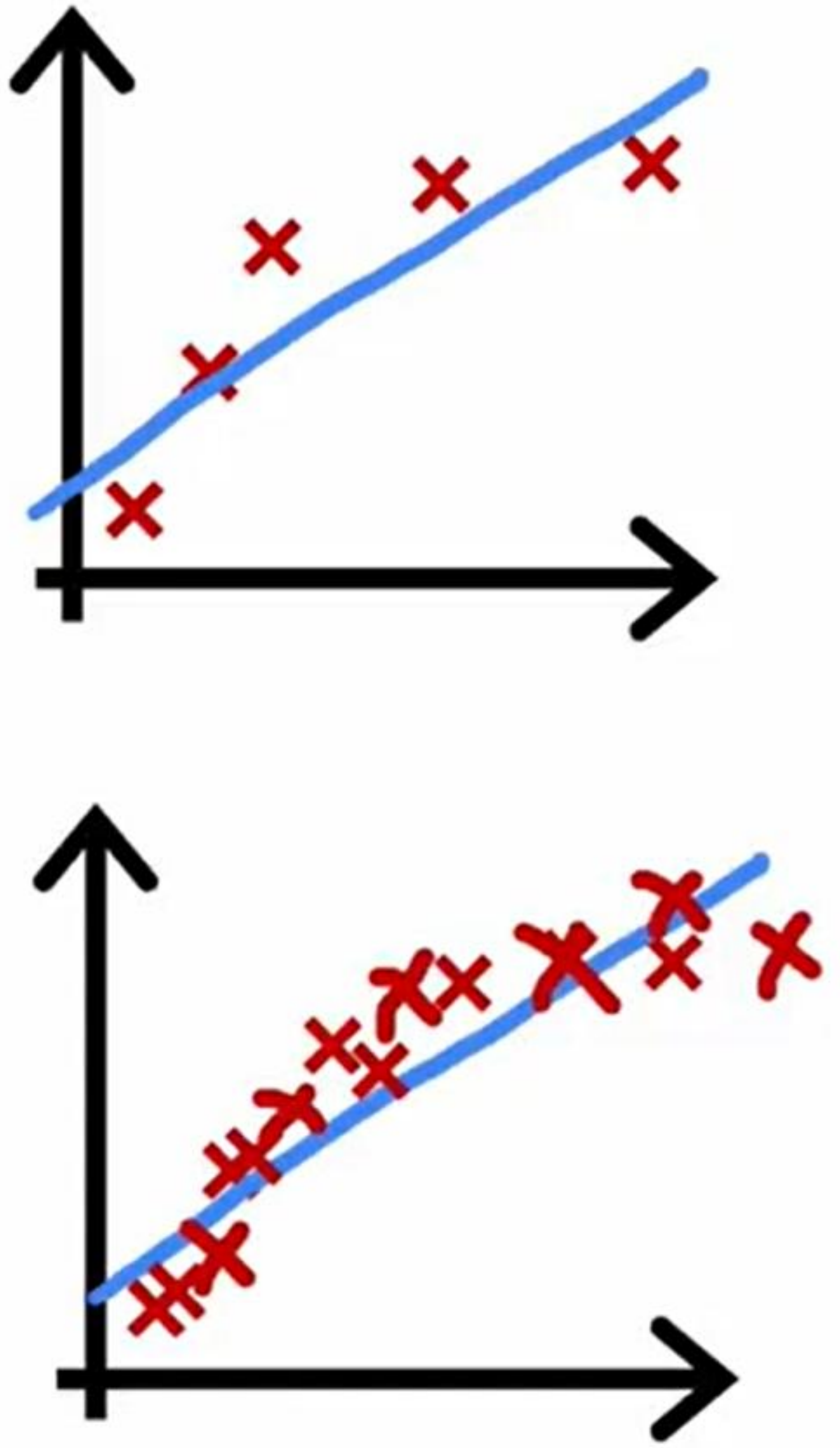
$f_{\vec{w},b}(x) = w_1 x + w_2 x^2 + b$



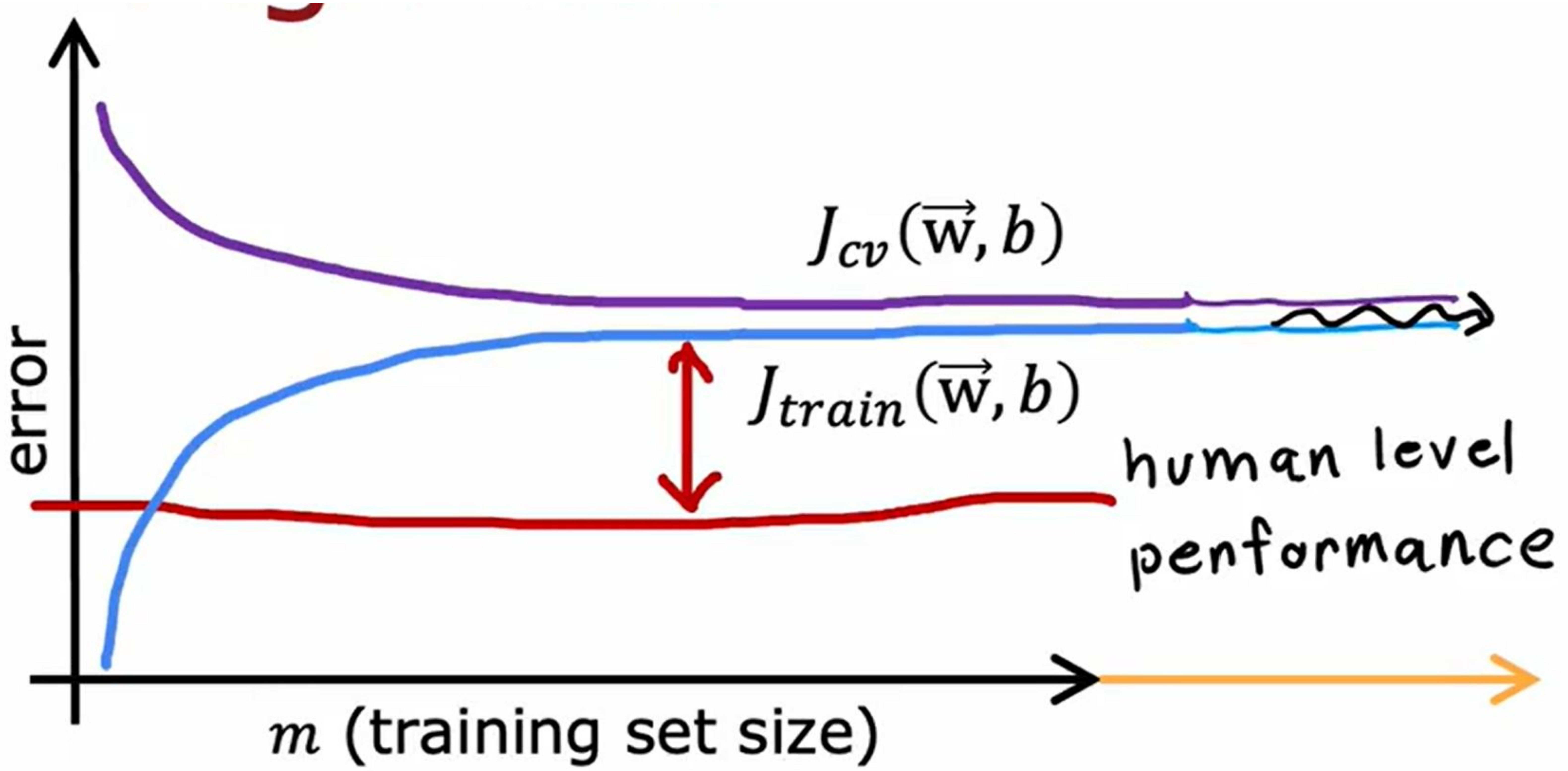
High bias



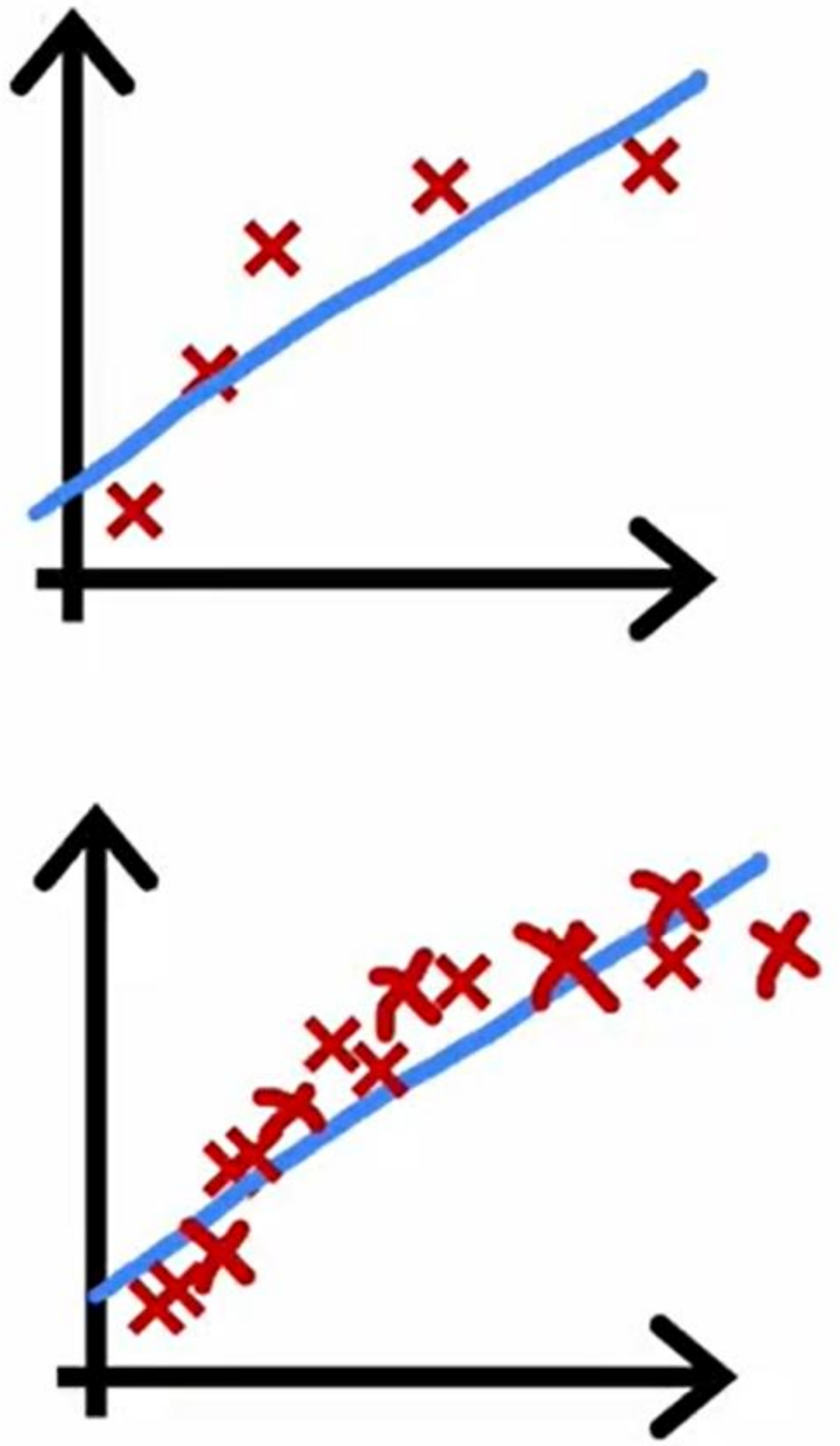
$$f_{\vec{\mathrm{w}},b}(x) = w_1 x + b$$



High bias



$f_{\vec{w},b}(x) = w_1x + b$

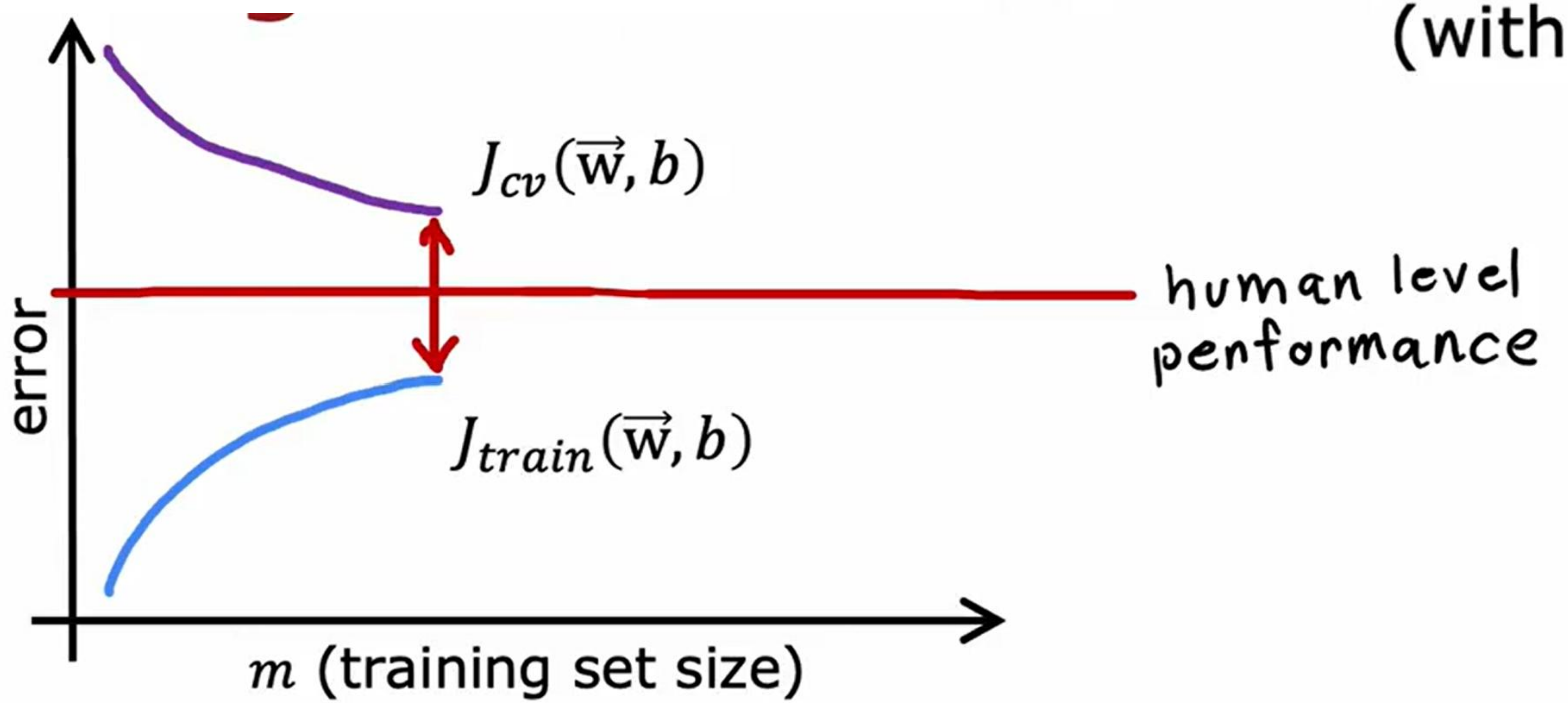


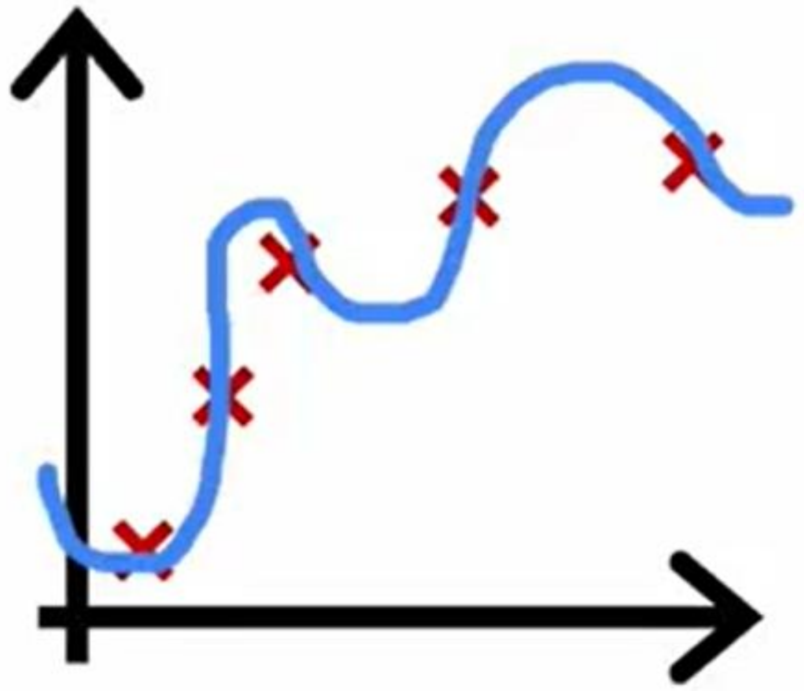
if a learning algorithm suffers from high bias, getting more training data will not (by itself) help much.

High variance

$$f_{\vec{w},b}(x) = w_1x + w_2x^2 + w_3x^3 + w_4x^4 + b$$

(with small λ)

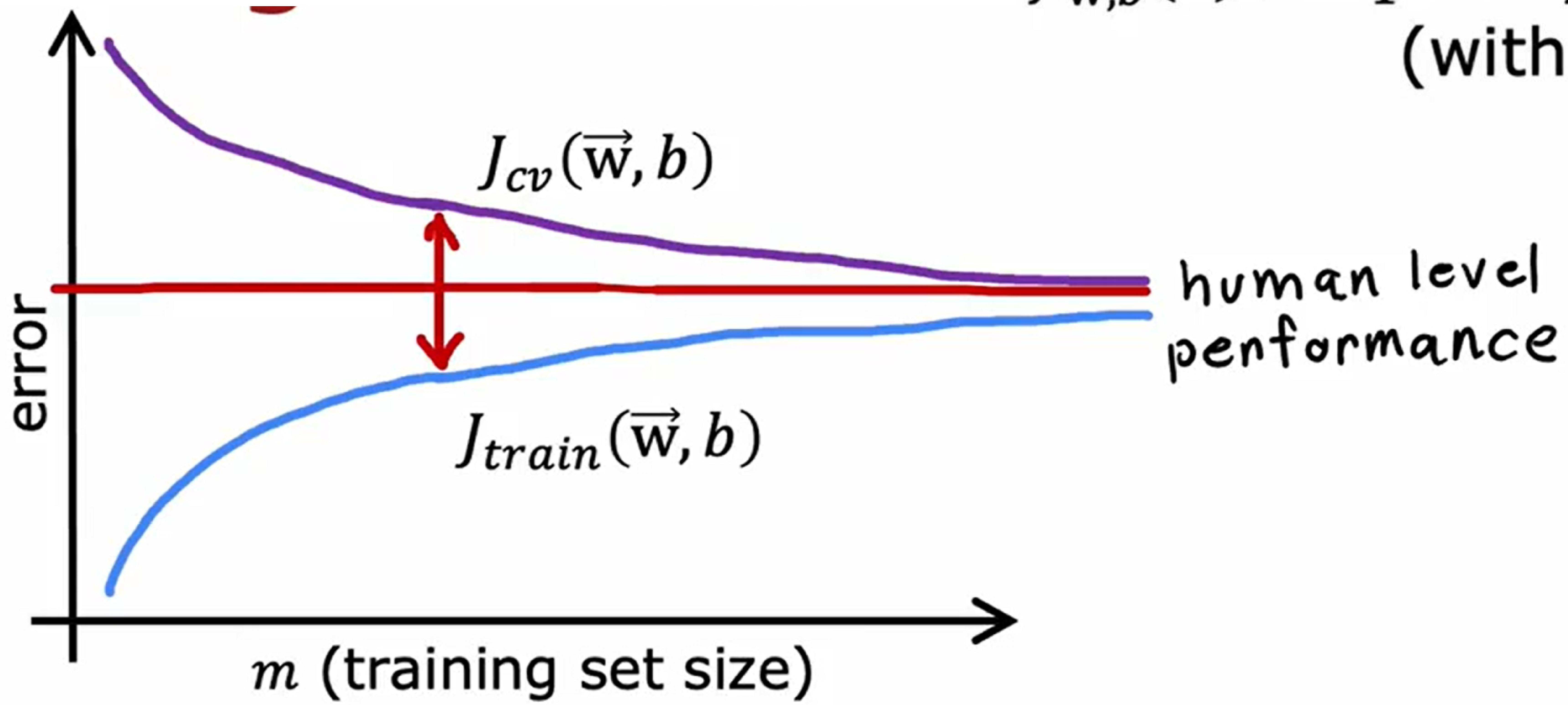


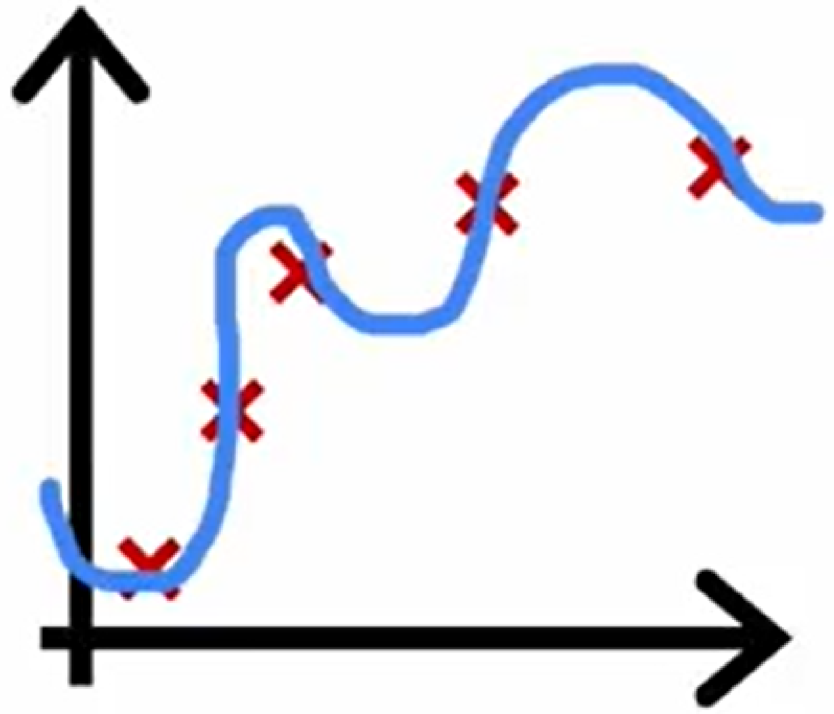


High variance

$$f_{\vec{w},b}(x) = w_1 x + w_2 x^2 + w_3 x^3 + w_4 x^4 + b$$

(with small λ)





If a learning algorithm suffers from high variance, getting more training data is likely to help.

Debugging a learning algorithm

You've implemented regularized linear regression on housing prices

$$J(\vec{\mathrm{w}},b)=\frac{1}{2m}\sum_{i=1}^{m}\left(f_{\vec{\mathrm{w}},b}\left(\vec{\mathrm{x}}^{(i)}\right)-y^{(i)}\right)^2+\frac{\lambda}{2m}\sum_{j=1}^{n}w_j^2$$

But it makes unacceptably large errors in predictions. What do you try next?

Get more training examples — fixes high variance
Try smaller sets of features
Try getting additional features
Try adding polynomial features $(x_1^2, x_2^2, x_1x_2, etc)$
Try decreasing λ
Try increasing λ

Debugging a learning algorithm

You've implemented regularized linear regression on housing prices

$$J(\vec{\mathrm{w}}, b) = \underbrace{\frac{1}{2m}\sum_{i=1}^{m}\left(f_{\vec{\mathrm{w}},b}\left(\vec{\mathrm{x}}^{(i)}\right) - y^{(i)}\right)^2}_{} + \underbrace{\frac{\lambda}{2m}\sum_{j=1}^{n} w_j^2}_{}$$

But it makes unacceptably large errors in predictions. What do you try next?

→ Get more training examples	fixes high variance
→ Try smaller sets of features $x, x^2, \cancel{x^3}, \cancel{x^4}, \cancel{x^5} \ldots$	fixes high variance
→ Try getting additional features	fixes high bias
→ Try adding polynomial features $(\underline{x_1^2}, \underline{x_2^2}, \underline{x_1 x_2}, etc)$	fixes high bias
→ Try decreasing λ	fixes high bias
→ Try increasing λ	fixes high variance

The bias variance tradeoff

$f_{\vec{w},b}(x) = w_1x + b$

$f_{\vec{w},b}(x) = w_1x + w_2x^2 + b$

$f_{\vec{w},b}(x) = w_1x + w_2x^2 + w_3x^3 + w_4x^4 + b$

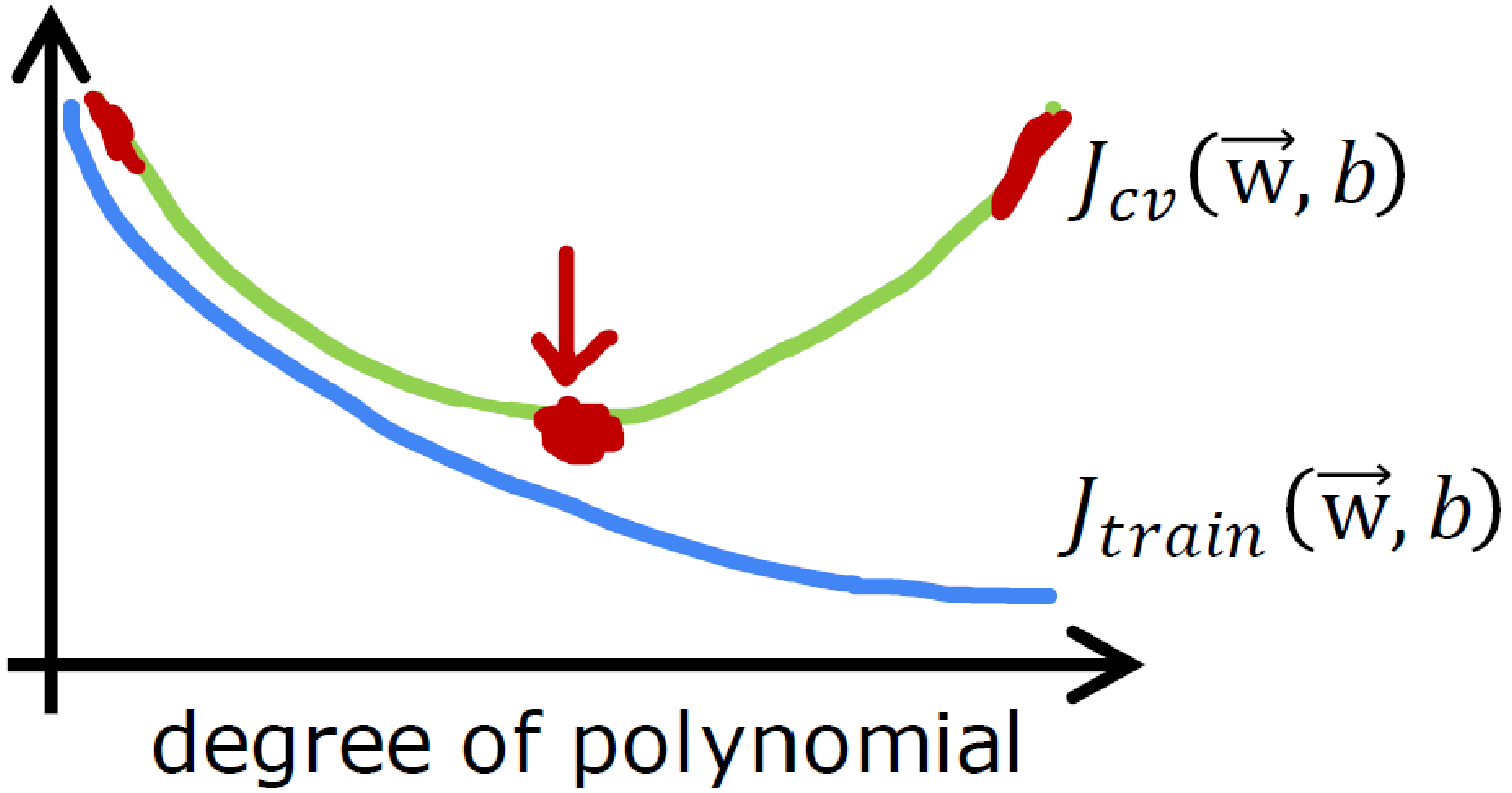
Simple model

High bias

tradeoff

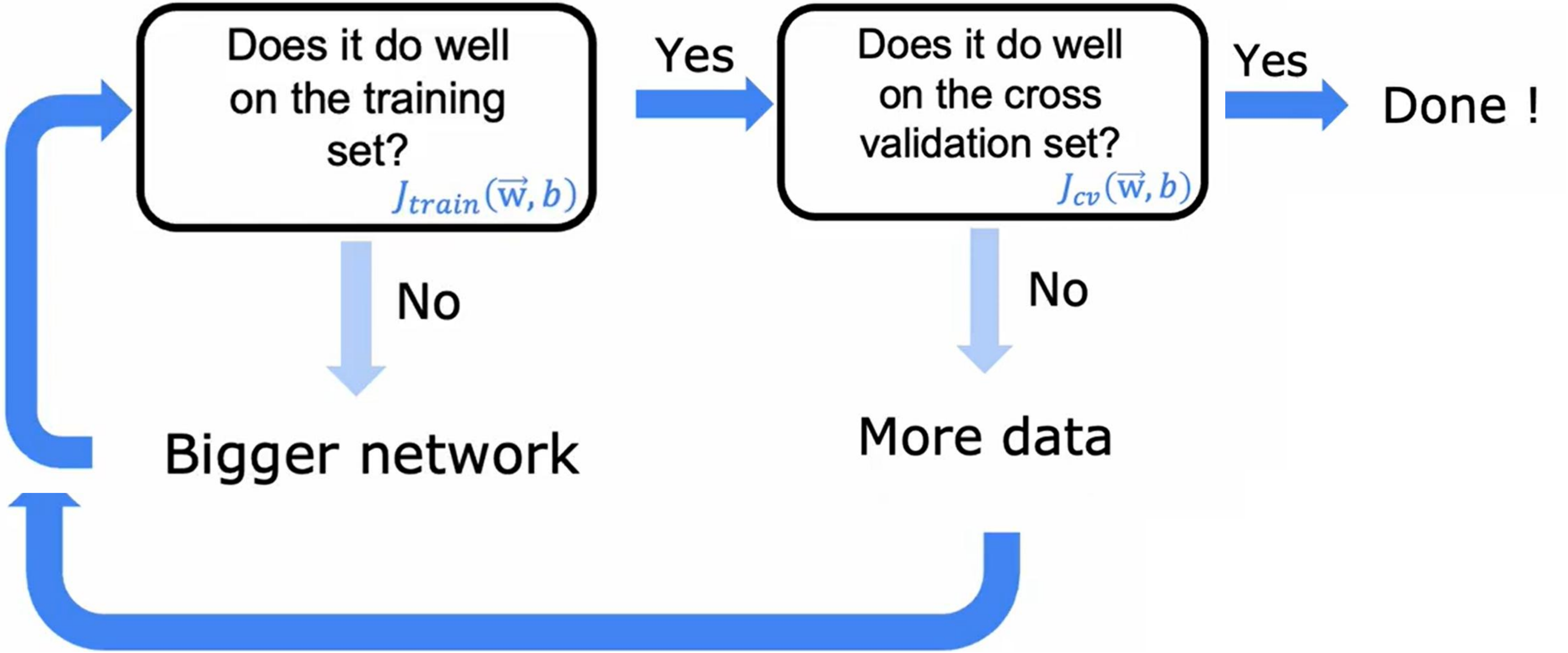
Complex model

High variance

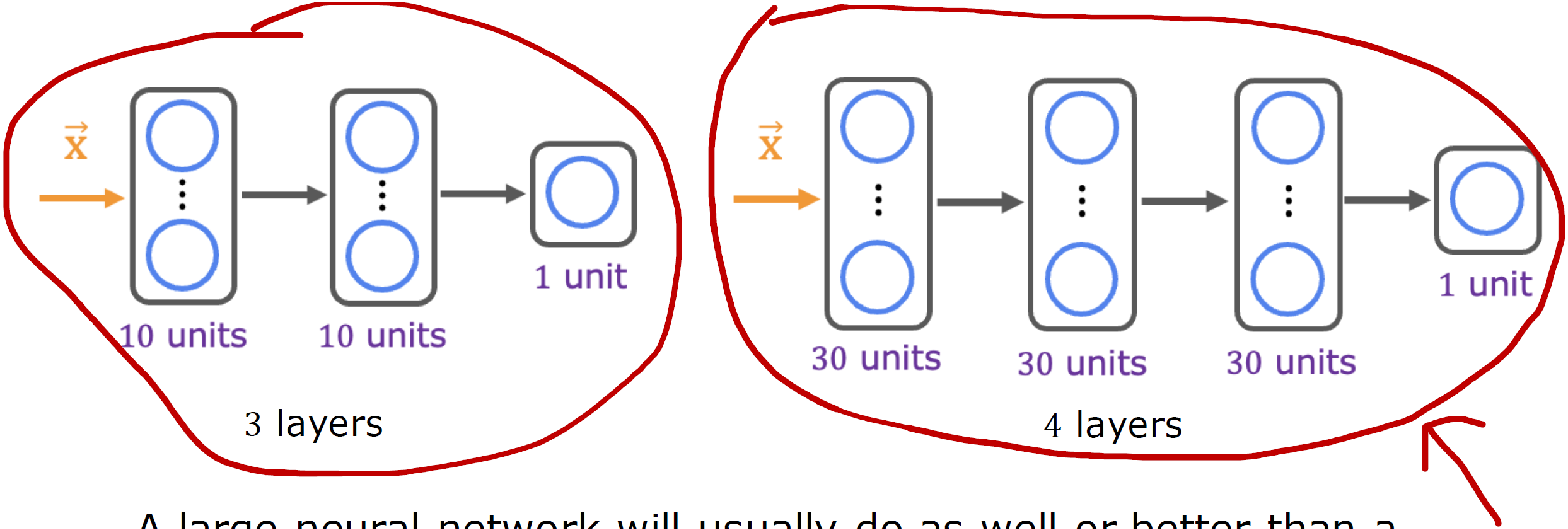


Neural networks and bias variance

Large neural networks are low bias machines



Neural networks and regularization



A large neural network will usually do as well or better than a smaller one so long as regularization is chosen appropriately.

Neural network regularization

$$J(\mathbf{W},\mathbf{B}) = \frac{1}{m}\sum_{i=1}^{m} \underbrace{L(f(\vec{\mathrm{x}}^{(i)}), y^{(i)})} + \frac{\lambda}{2m}\underbrace{\sum_{all\ weights\ \mathbf{W}}}(w^2)$$

b

Unregularized MNIST model

```
layer_1 = Dense(units=25, activation="relu")
layer_2 = Dense(units=15, activation="relu")
layer_3 = Dense(units=1, activation="sigmoid")
model = Sequential([layer_1, layer_2, layer_3])
```

Regularized MNIST model

λ

```
layer_1 = Dense(units=25, activation="relu", kernel_regularizer=L2(0.01))
layer_2 = Dense(units=15, activation="relu", kernel_regularizer=L2(0.01))
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```
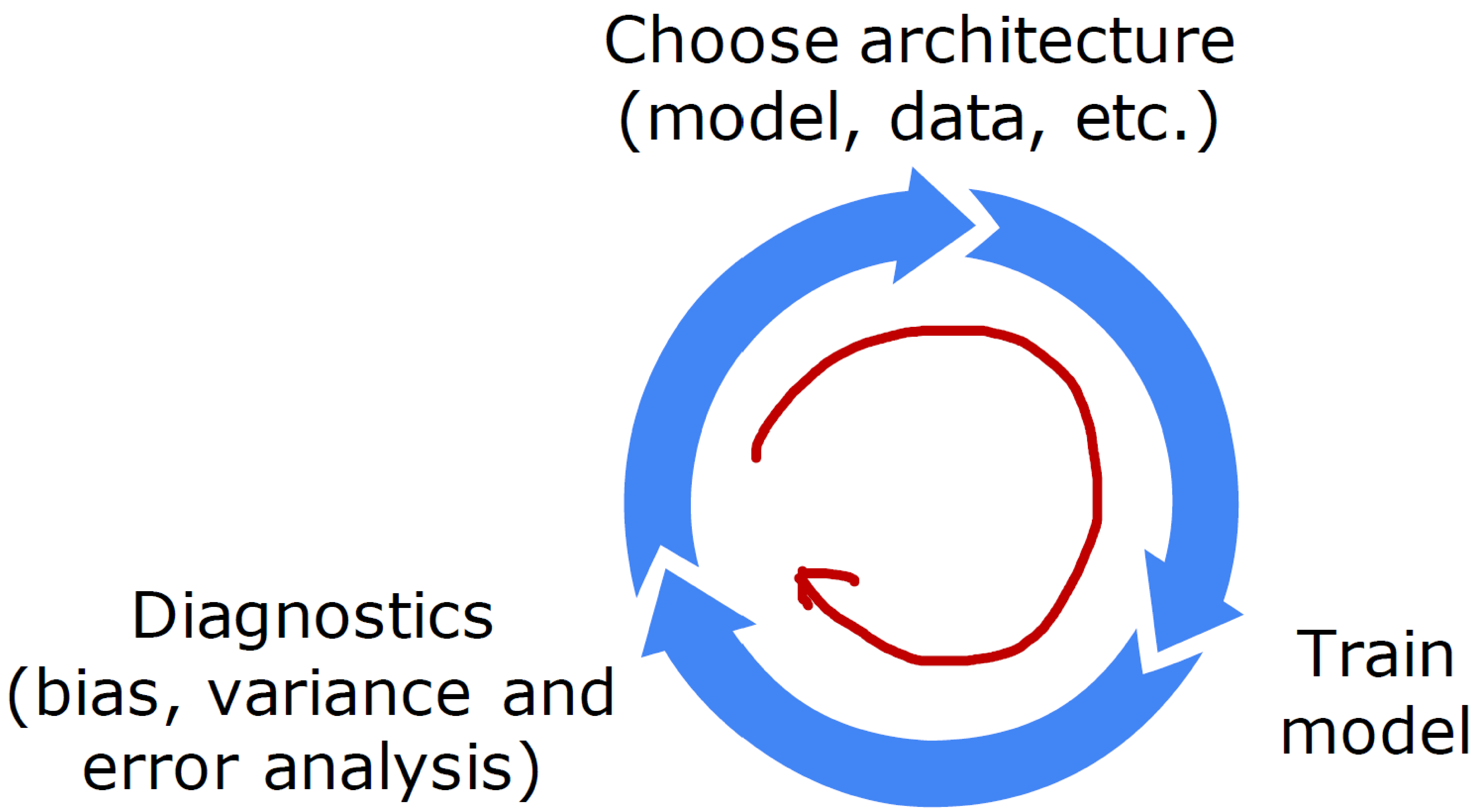

Iterative loop of ML development



Spam classification example

From: cheapsales@buystufffromme.com
To: Andrew Ng
Subject: Buy now!

Deal of the week! Buy now!
Rolex w4tchs - $100
Med1cine (any kind) - £50
Also low cost M0rgages
available.

From: Alfred Ng
To: Andrew Ng
Subject: Christmas dates?

Hey Andrew,
Was talking to Mom about plans
for Xmas. When do you get off
work. Meet Dec 22?
Alf

Building a spam classifier

Supervised learning: $\vec{x}$ = features of email

y = spam (1) or not spam (0)

Features: list the top 10,000 words to compute $x_1, x_2, \cdots, x_{10,000}$

$$\vec{x} = \begin{array}{c} \left[\begin{array}{c} 0 \\ 1 \\ \cancel{2}\ \cancel{2}\ 1 \\ 1 \\ 0 \\ \vdots \end{array}\right] \end{array} \begin{array}{l} \text{a} \\ \text{andrew} \\ \text{buy} \\ \text{deal} \\ \text{discount} \\ \vdots \end{array}$$

From: cheapsales@buystufffromme.com
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Building a spam classifier

How to try to reduce your spam classifier's error?

- Collect more data. E.g., "Honeypot" project.
- Develop sophisticated features based on email routing (from email header).
- Define sophisticated features from email body. E.g., should "discounting" and "discount" be treated as the same word.
- Design algorithms to detect misspellings. E.g., w4tches, med1cine, m0rtgage.

...

Error analysis

m_{cv} = 500 examples in cross validation set.

Algorithm misclassifies 100 of them.

Manually examine 100 examples and categorize them based on common traits.

- Pharma: 21
- Deliberate misspellings (w4tches, med1cine): 3
- Unusual email routing: 7
- Steal passwords (phishing): 18
- Spam message in embedded image: 5

Error analysis

m_{cv} = ~~500~~ 5000 examples in cross validation set.

Algorithm misclassifies ~~100~~ 1000 of them.

Manually examine 100 examples and categorize them based on common traits.

- Pharma: 21 → more data features
- Deliberate misspellings (w4tches, med1cine): 3
- Unusual email routing: 7
- Steal passwords (phishing): 18 → more data features
- Spam message in embedded image: 5

Building a spam classifier

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...

Data Augmentation

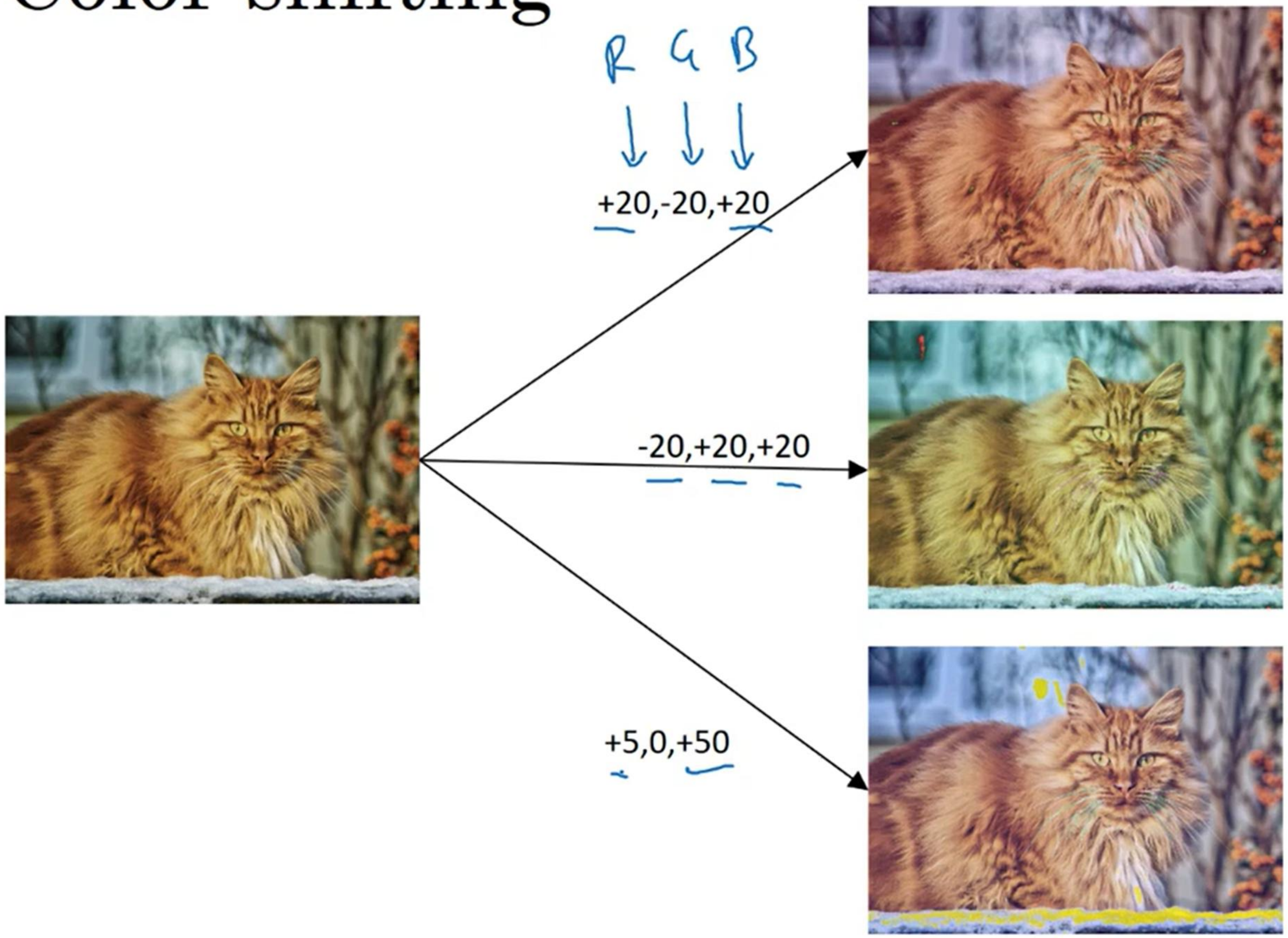
Data augmentation is a technique used to artificially increase the size of a training dataset by applying various transformations to the existing data.

Why Use Data Augmentation?

- Increased Dataset Size
- Regularization
- Improved Generalization

Data Augmentation

Color shifting

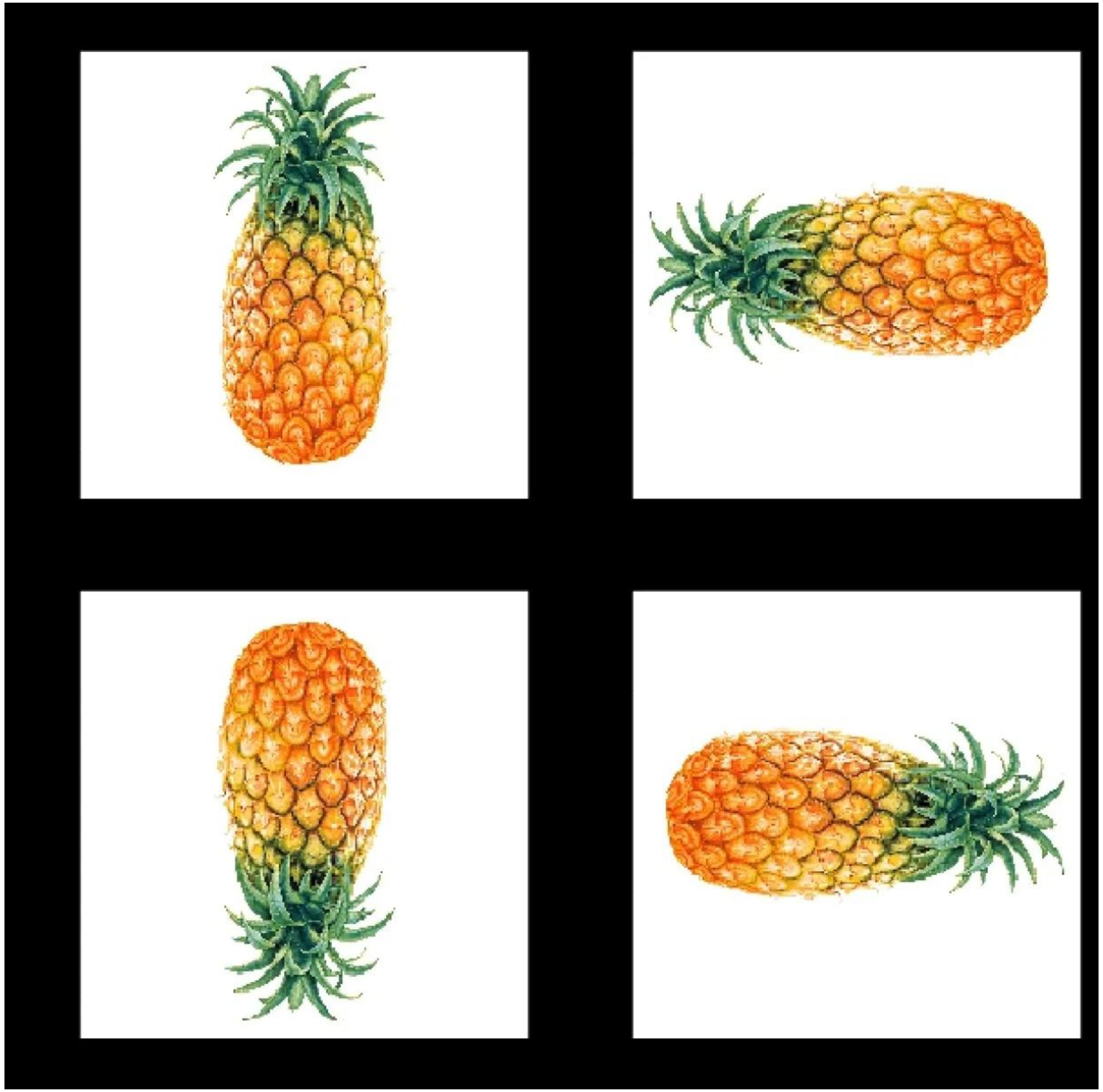


Adding noise

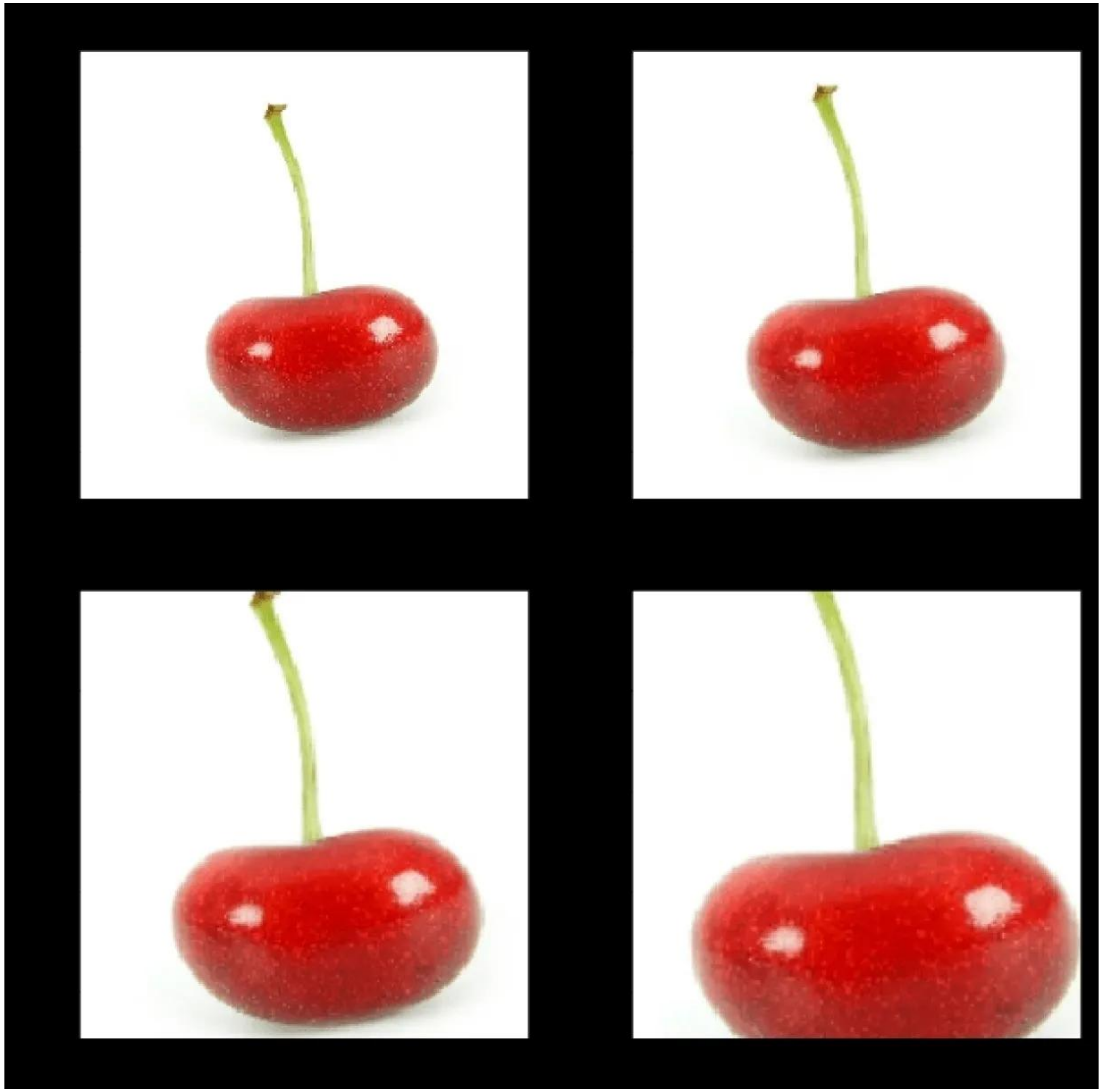
Cropping

Flipping

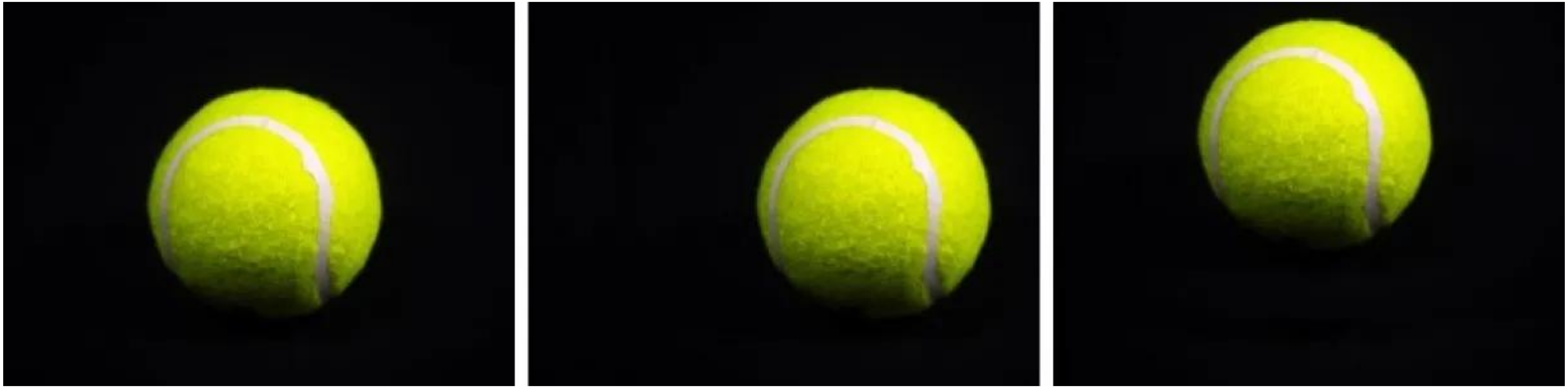
Rotation



Scaling

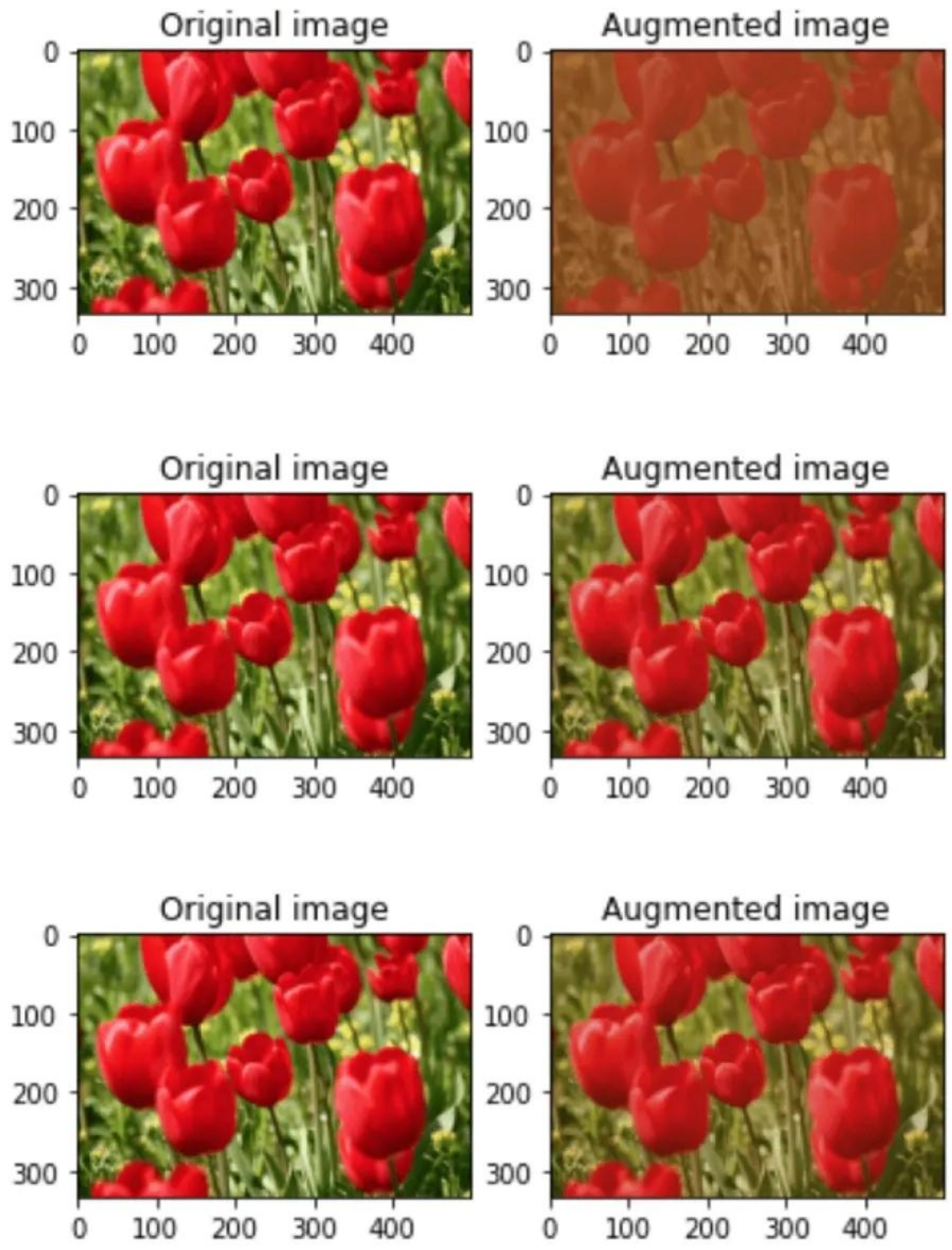


Translation



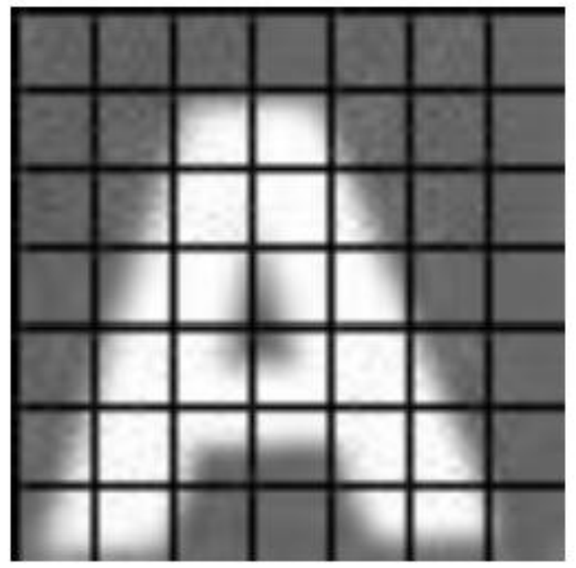
Brightness

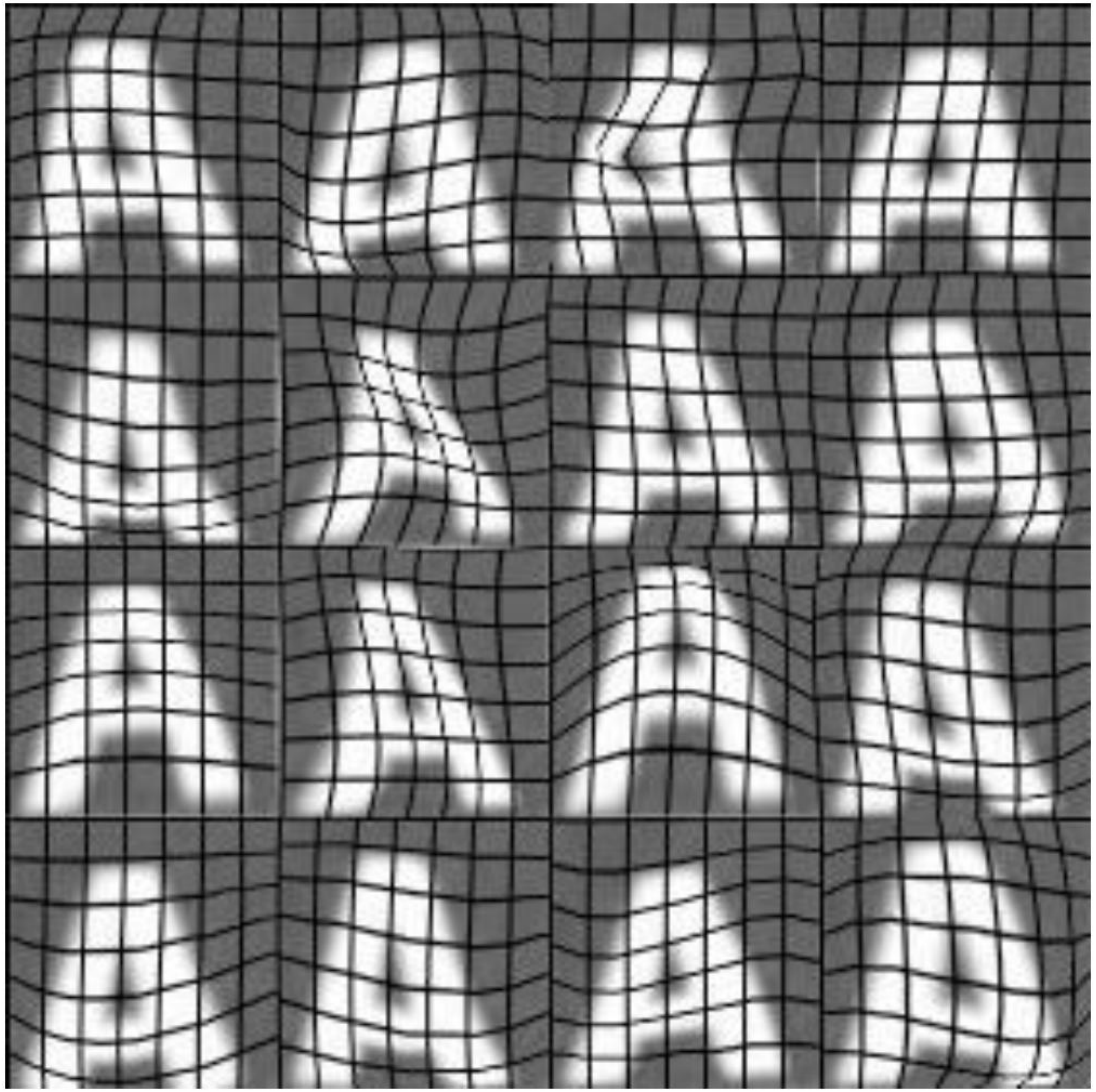
Contrast


Original image
Augmented image
Original image
Augmented image
Original image
Augmented image
0
100
200
300
400

Saturation

Distortions





Data augmentation for speech

Speech recognition example

Original audio (voice search: "What is today's weather?")

+ Noisy background: Crowd

+ Noisy background: Car

+ Audio on bad cellphone connection

Data synthesis

Synthesis: using artificial data inputs to create a new training example

Artificial data synthesis for photo OCR

Real data

Artificial data synthesis for photo OCR

Real data

Synthetic data

Engineering the data used by your system

Conventional model-centric approach:

AI = Code + Data
(algorithm/model)

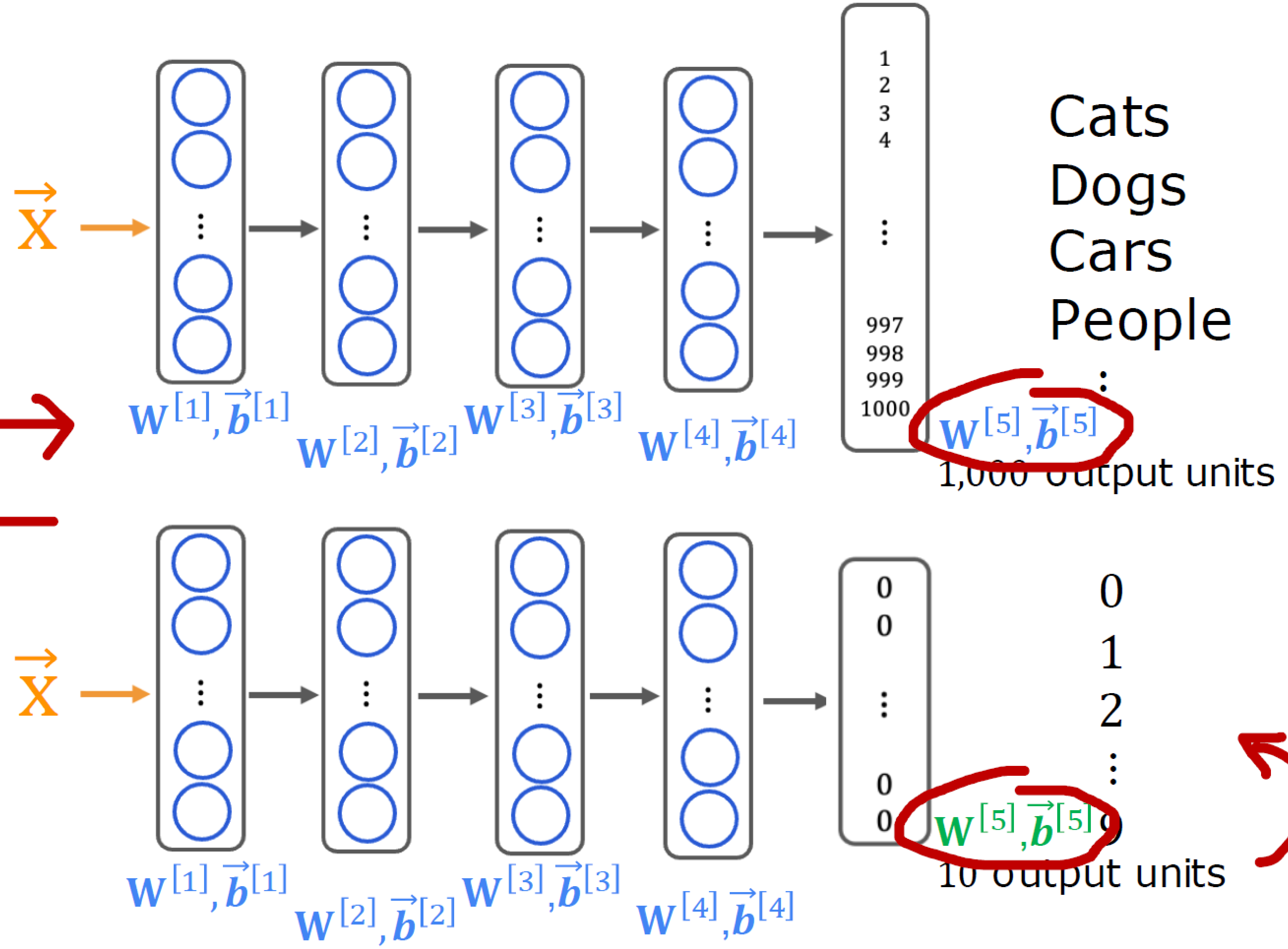
Work on this

Data-centric approach:

AI = Code + Data
(algorithm/model)

Work on this

Transfer Learning



1,000 classes

0, 1, 2, ... ,9

1 million images

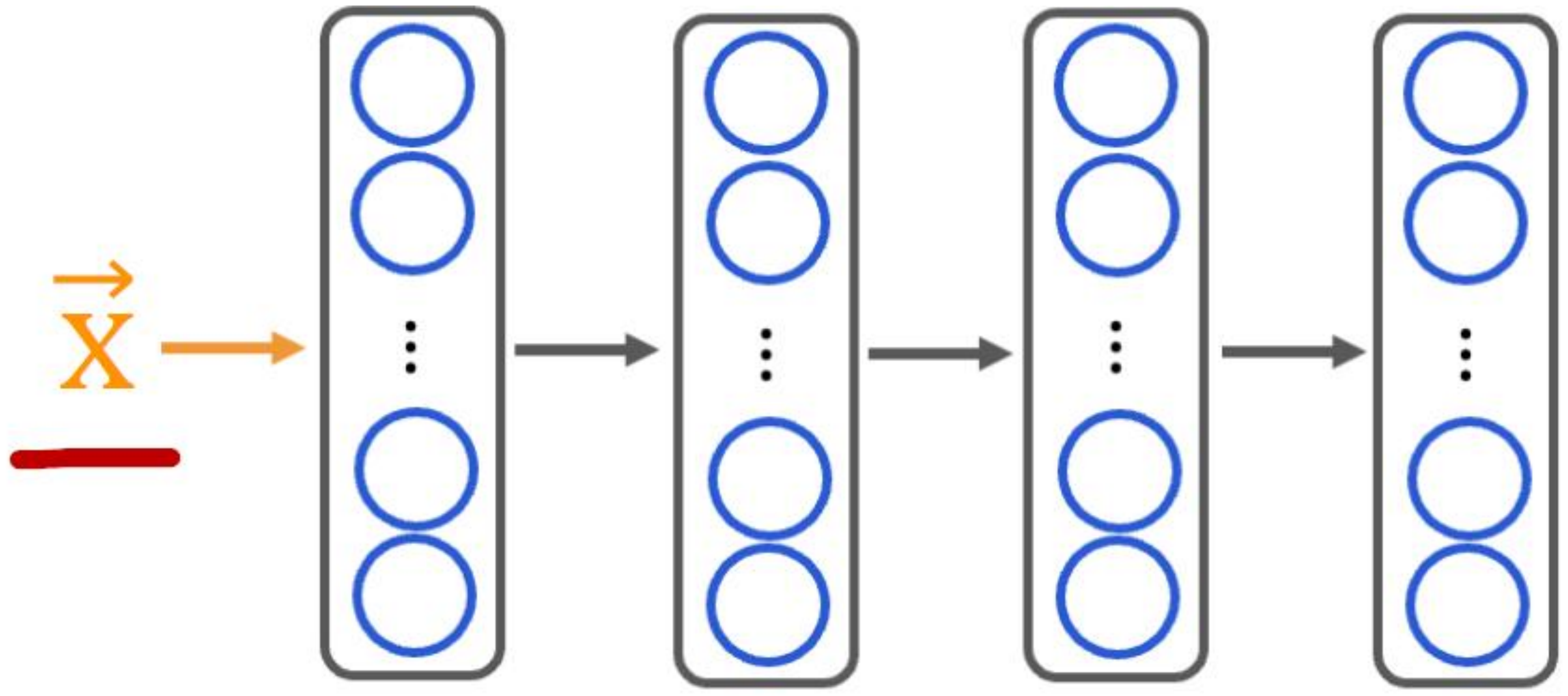
Supervised pretraining

Fine tuning

Option 1: only train output layers parameters.

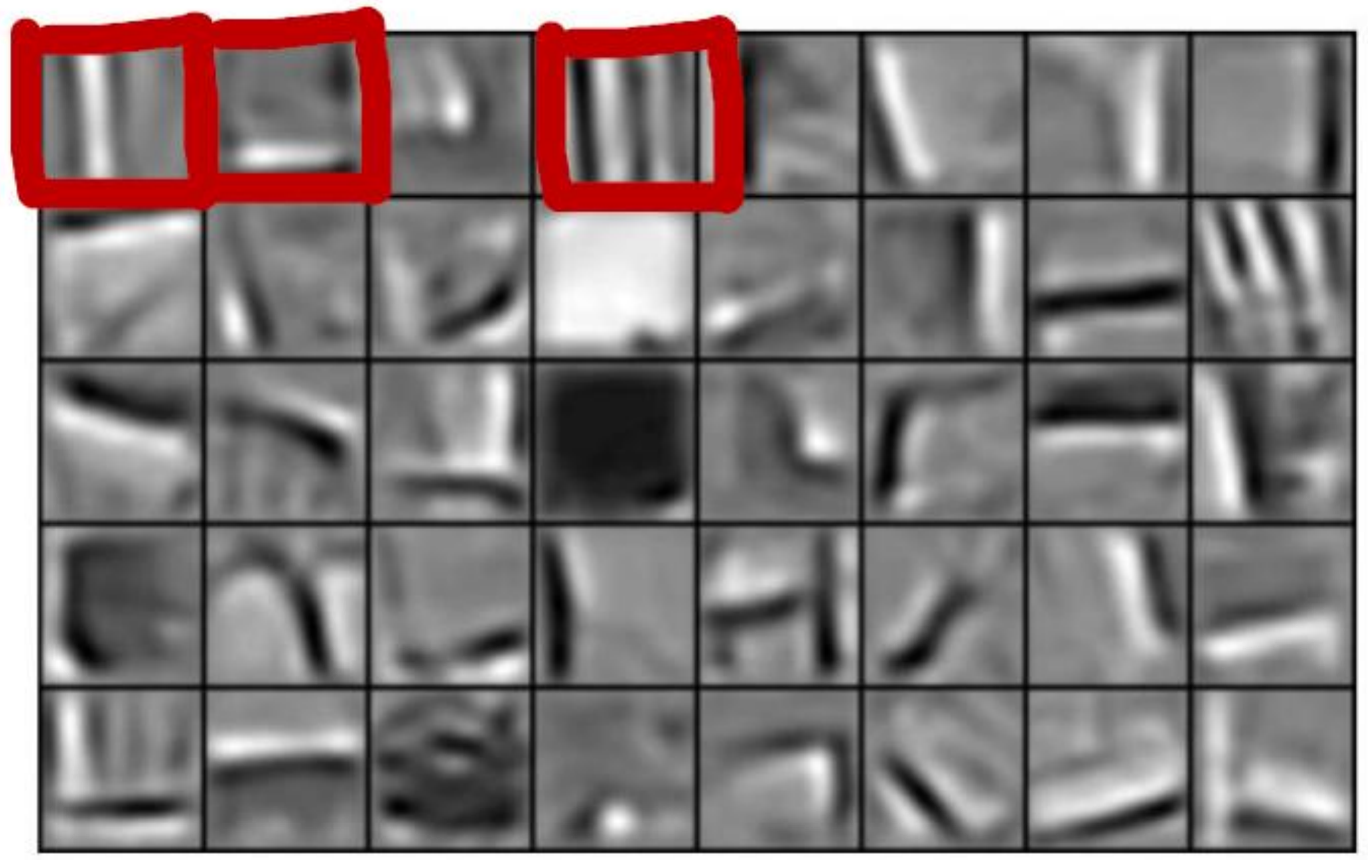
Option 2: train all parameters.

Why does transfer learning work?

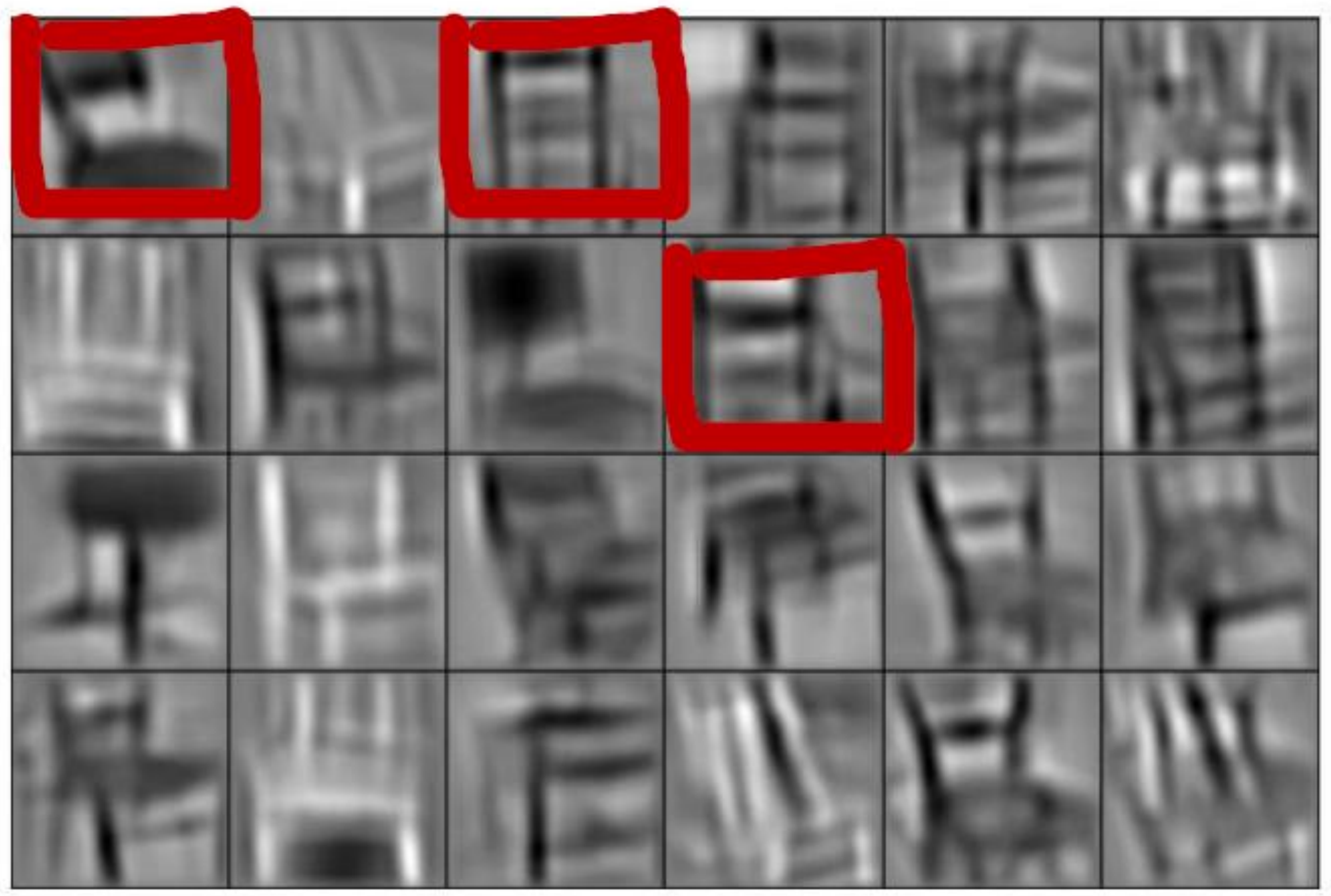


use the same input type

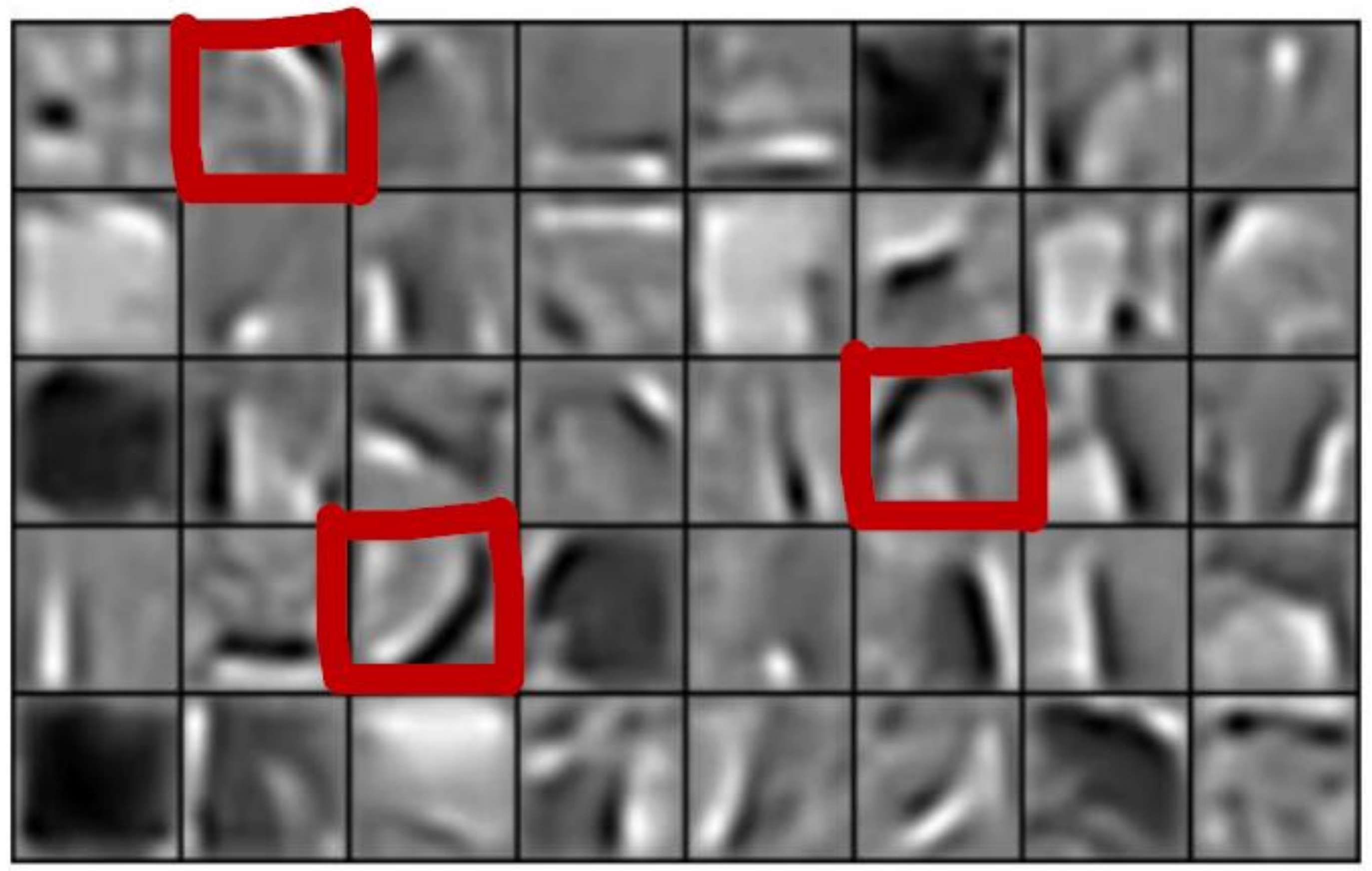
detects edges | detects corners | detects curves/basic shapes



Edges



Corners



Curves / basic shapes

Transfer learning summary

1. Download neural network parameters pretrained on a large dataset with same input type (e.g., images, audio, text) as your application (or train your own).

1M

2. Further train (fine tune) the network on your own data.

1000

50

Full cycle of a machine learning project

Fairness, bias, and ethics (Bias)

Hiring tool that discriminates against women.

Facial recognition system matching dark skinned individuals to criminal mugshots.

Biased bank loan approvals.

Adverse use cases

Deepfakes

Spreading toxic/incendiary speech through optimizing for engagement.

Generating fake content for commercial or political purposes.

Using ML to build harmful products, commit fraud etc.

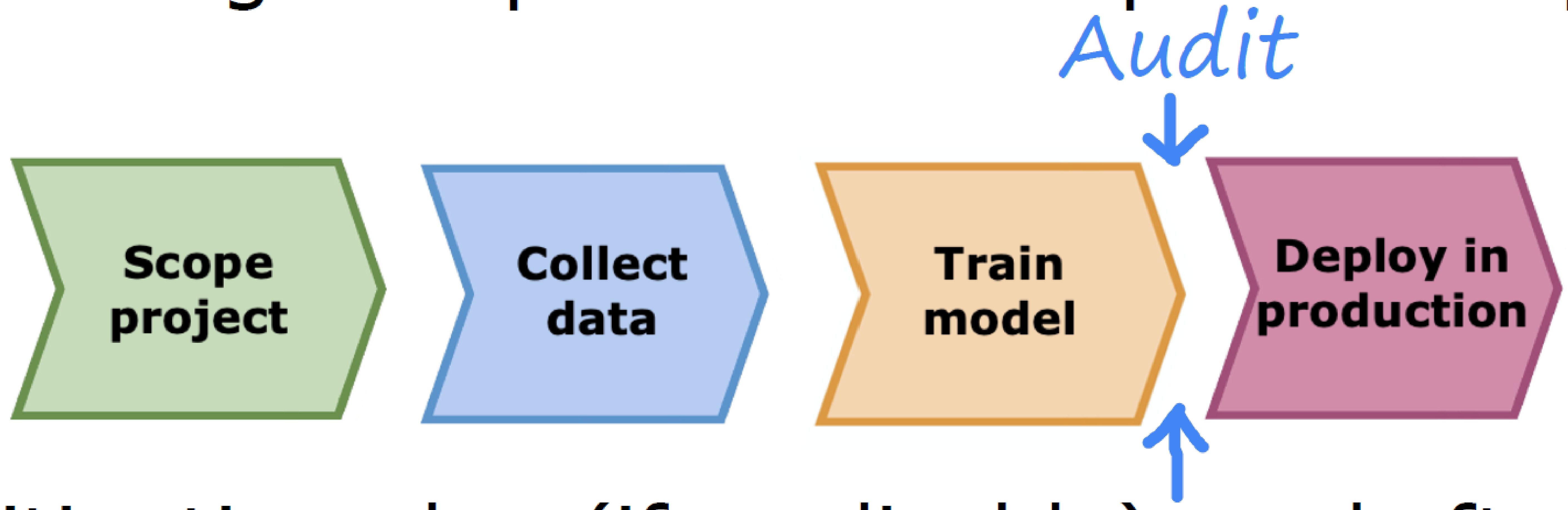
Spam vs anti-spam : fraud vs anti-fraud.

Guidelines

Get a diverse team to brainstorm things that might go wrong, with emphasis on possible harm to vulnerable groups.

Carry out literature search on standards/guidelines for your industry.

Audit systems against possible harm prior to deployment.



Develop mitigation plan (if applicable), and after deployment, monitor for possible harm.

Thank You

Any questions?

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Source & Special Thanks to Andrew Ng (Coursera)
Machine Learning Course 2

https://www.youtube.com/@Deeplearningai

https://www.youtube.com/watch?v=ggWLvh484hs&list=PLyoNSC4BT4eVpykPF0Yx8C1Zs50XtD17L

https://research.aimultiple.com/data-augmentation-techniques/